RDIMS #1230825

QUALIFICATION STANDARD

AND PLAN (QSP)

USQ – Roundspersons HARRY DEWOLF Class

NQual (AMBT)

MITE ID

Residential Phase – Nil

OJT – 122996

RDIMS #1230825

Regular Force: Marine Technician (00379)

Primary Reserve: Marine Technician – Reserve (00293-01)

Training Authority (TA): Commander Naval Personnel and Training Group (CNTG)

Curriculum Control Authority (CCA): NFS(P)

RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0 (QSWB Month Year)			
			NTG HQ
1.01			CCA
1.1		10 May 2024	CNTG

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Training Group (CNTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the following references:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-003 Canadian Forces Officer General Specification dated 01 Jul 15;

   c. A-PD-055-002/PP-002 Non-Commissioned Member General Specification dated 2020-11-20;

   d. A-PD-055-002/PP-001 Name of Occupation Job Based Specification or Job Based Occupational Specification dated 2018-03-23; and

   e. NAVORD 4500-0 - Structured Delegation, and NTORD 4500-22 – Qualification Standard and Plan Production and Management.

3. This QSP publication is effective on receipt and is the first QSP for this position/platform. This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened 1 November 2022 at NFS(P) working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened 1 November 2022 at NFS(P) working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a Training Waiver in accordance with NTORD 4500-25 is required.

RDIMS #1230825

QUALIFICATION STANDARD

CHAPTER 1: – QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the Marine Technician Harry DeWolf Roundsperson to perform their duties as a Roundsperson on a Harry DeWolf Class vessel. Possession of this qualification will allow personnel to conduct on watch rounds of a Harry DeWolf Class vessel. The member will also be able to respond to engineering difficulties and emergencies as a Roundsperson.

GRANTING OF NQUAL

2. A trainee will achieve the qualification (AMBT) USQ HARRY DEWOLF Class when the standards listed in the Performance Objectives (POs) have been met.

CO-QUALIFICATION

3. There are no additional qualifications associated with this training.

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

   a. RQ-S3 Marine Technician MITE ID 122592.

CHAPTER 2: – PERFORMANCE OBJECTIVES

PO 001 PERFORM MAIN PROPULSION SYSTEM ROUNDS

1. Performance: Perform Main Propulsion System Rounds.

2. Conditions:

   a. Given:

      (1) HARRY DEWOLF class ship;
      (2) PPE;
      (3) Drawings;
      (4) Log books;
      (5) Supervision;
      (6) References; and
      (7) Test equipment.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any condition day or night, at sea or alongside.

3. Standard: The Marine Technician Basic Maintainer shall perform main propulsion system rounds IAW refs to include:

   a. Inspecting shafting and propeller systems by:

      (1) Operating shaft line turning gear locally;

      (2) Operating shaft line stern tube inflatable seal;

      (3) Operating shaft locking/braking device;

      (4) Inspecting the Propulsion Motors; and

      (5) Operating Quality water package system locally.

   b. Inspect Main Propulsion Converter to include:

      (1) Inspecting converter cooling cabinet;

      (2) Inspecting converter control cabinet; and

(3) Inspecting propulsion transform cooling fans.

4. Reference: A2, A7, A8, A9, A87, A90, A91, and A12.

5. Remarks:

a. References at the teaching point level will need to refined during the pilot serial.

PO 002 PERFORM PRIMARY PG&D ROUNDS

1. Performance: Perform Primary PG&D Rounds.

2. Conditions:

   a. Given:

      (1) HARRY DEWOLF class ship;
      (2) PPE;
      (3) Drawings;
      (4) Log books;
      (5) Supervision;
      (6) References; and
      (7) Test equipment.

   b. Denied: Nil.

   c. Environment: Any condition day or night, at sea or alongside.

3. Standard: The Marine Technician Basic Maintainer shall perform primary PG&D rounds IAW refs by:

   a. Inspecting combustion engine by:

      (1) Inspecting the main diesel engine; and

      (2) Starting up/Shutting down MAN diesel engine (with assistance).

   b. Inspecting Exhaust Systems, to include:

      (1) Inspecting MAN Diesel Exhaust;

      (2) Inspecting Incinerator Exhaust;

      (3) Inspecting Water Heater Exhaust;

      (4) Inspecting EHDG Exhaust:

      (5) Monitor Selective Catalytic Reactor (SCR), to include:

         (a) UREA Injection Stop Mode.

4. Reference: A2, A3, A15, A20, A22, A23, A33, A34, and A35.

5. Remarks:

a. References at the teaching point level will need to refined during the pilot serial.

PO 003 PERFORM MARINE ENGINEERING AUXILIARY ROUNDS

1. Performance: Perform Marine Engineering Auxiliary Rounds.

2. Conditions:

   a. Given:

      (1) HARRY DEWOLF class ship;
      (2) PPE;
      (3) Drawings;
      (4) Log books;
      (5) Supervision;
      (6) References; and
      (7) Test equipment.

   b. Denied: Nil.

   c. Environment: Any condition day or night, at sea or alongside.

3. Standard: The Marine Technician Basic Maintainer shall perform marine engineering auxiliary rounds IAW refs to include:

   a. Inspecting steering systems to include:

      (1) Operating steering systems manually.

   b. Inspecting Bow Thruster to include:

      (1) Conducting per-start checks;

      (2) Operating the Bow Thruster locally; and

      (3) Emergency and Abnormal operating procedures

   c. Inspecting FOS by:

      (1) Inspecting FOS when in operation; and

      (2) Operating FOS locally, to include:

         (a) Operating FO pumps locally; and

         (b) Operating FOS filter systems

   d. Inspecting Fuel Oil Transfer (FOT) system by:

(1) Inspecting FOT system when in operation; and
(2) Operating FOT locally to include:
(a) Operating FOT pumps locally.

e. Inspect Fuel/Oil Purifiers by:

(1) Inspecting Fuel/Oil Purifiers while in operation; and
(2) Operating Fuel/Oil Purifiers locally.

f. Inspecting diesel lube oil filling & transfer system by:

(1) Operating lube oil purifier locally; and
(2) Operating waste oil pump locally.

g. Inspecting compressed air systems by:

(1) Operating Ship's Start Air system locally;
(2) Operating Service Air system locally; and
(3) Operating Breathing Air Compressors locally.

h. Inspecting fresh water system by:

(1) Inspecting visually, fresh water systems;
(2) Operating fresh water systems, locally;
(3) Inspecting visually, potable water purification; (chlorine) dousing system; and
(4) Operating reverses osmosis desalination (ROD) unit.

i. Inspecting black and grey water collection/treatment systems to include:

(1) Inspecting black water system; and
(2) Inspecting grey water system.

j. Inspecting Oil Pollution and Abatement system by;

(1) Inspecting Grease Trap;
(2) Operating the Oily Water Separator;
(3) Operating locally the Mobile Separator; and
(4) Operating locally all pumps.

k. Inspecting Ballast Water Treatment System by:

(1) Inspecting visually, BWTS;
(2) Conducting pre-start checks; and
(3) Operating locally.

l. Inspecting the Ballast Water system by:

(1) Operating the Ballast Water system locally.

m. Inspecting Incinerator System by:

(1) Inspecting visually the Incinerator;
(2) Emergency stop procedures; and
(3) Operating locally.

n. Inspecting Equipment Cooling systems by:

(1) Inspecting Central Fresh Water Cooling System; and
(2) Inspecting Seawater Feed/Cooling System.

o. Inspecting Thermal Oil system by:

(1) Operating the Thermal Oil System.

p. Inspecting Tank Heating System by:

(1) Operating Circulation pumps in local.

q. Inspect Stability systems to include:

(1) Inspecting Active Roll Fin Stabilization system by:

(a) Inspecting visually, hydraulic system;
(b) Conducting pre-start checks; and
(c) Emergency fin housing operation.

(2) Inspecting Heeling System to include:

(a) Conducting an emergency stop; and
(b) Operating locally.

4. Reference: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83.

5. Remarks:

a. References at the teaching point level will need to refined during the pilot serial.

RDIMS #1230825

PO 004 PERFORM SECONDARY PG&D ROUNDS

1. Performance: Perform Secondary PG&D Rounds.

2. Conditions:

   a. Given:

      (1) HARRY DEWOLF class ship;
      (2) PPE;
      (3) Drawings;
      (4) Log books;
      (5) Supervision;
      (6) References; and
      (7) Test equipment and applicable tools.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Basic Maintainer shall perform secondary PG&D rounds IAW refs:

   a. Inspecting Secondary PG&D by:

      (1) Inspecting Secondary PG&D systems to include:

         (a) Operating LV power generation systems remotely; and

         (b) Operating 440V distribution system.

      (2) Inspecting electrical LV switchboard remotely to include:

         (a) Operating LV and Emergency SWBD locally;

         (b) Paralleling electrical loads remotely; and

         (c) Operating switchboard breakers remotely.

      (3) Testing UPS to include:

         (a) Inspecting uninterruptable power supplies wiring harness.

         (b) Testing switching unit auto to include:

i. Testing switching unit manual.

b. Inspecting electrical systems by:

(1) Inspecting whistle assemblies to include:

(a) Testing ship’s whistle;

(b) Testing whistle speed controllers; and

(2) Inspecting whistle at will switches to include:

(a) Testing whistle at all switches.

(3) Operate IPMS alarm monitoring system to include:

(a) Testing alarm relay outputs;

(b) Testing Dead Man Alarm System;

(c) Testing Noxious Gas Detection System; and

(d) Testing man locked in fridge alarm system.

(4) Operating Nav lights locally.

c. Inspecting cathodic protection system by:

(1) Operating cathodic protection system to include:

(a) Conducting hull potential test;

(b) Inspecting shaft grounding system; and

(c) Repairing sacrificial anode.

(2) Operating portable reference electrode; and

(3) Recording cathodic protection readings.

d. Inspecting Batteries by:

(1) Charging Batteries to include:

(a) Verifying battery state;

(b) Recording electrolyte specific gravity readings;

(c) Topping up electrolyte levels; and

(d) Cleaning batteries.

(2) Replacing Batteries;

(3) Updating battery log; and

(4) Securing battery related HAZMAT by;

(a) Applying WHMIS guidelines

e. Inspecting small boat electrics by:

(1) Completing a function test.

f. Operating Deck Machinery by:

(1) Operating the winch power in all modes; and

(2) Operating RAS winches.

g. Conducting electrical pre sail checks by:

(1) Verifying cathodic protection system;

(2) Verifying steering system to include:

(a) Flashing up and shutting down of steering system; and

(b) Changing over from primary to secondary steering.

(3) Testing alarm systems to include:

(a) Testing fog horn/whistles;

(b) Testing Dead Man Alarm System;

(c) Testing crash on deck alarm; and

(d) Testing man locked in fridge alarm.

(4) Inspecting battery condition;

(5) Testing small boat electrics;

(6) Testing damage control consoles to include testing DC alarm systems

(7) Operating deck machinery;

(8) Verifying navigation lighting; and

(9) Verifying status of power generation and distribution equipment.

h. Inspecting the Emergency Harbour Diesel Generator (EHDG) to include:

(1) Operating the EHDG.

4. Reference: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42.

5. Remarks:

a. References at the teaching point level will need to refined during the pilot serial.

PO 005 PERFORM DECK AND HULL EQUIPMENT ROUNDS

1. Performance: Perform Deck and Hull Equipment Rounds

2. Conditions:

    a. Given:

        (1) HARRY DEWOLF class ship;

        (2) PPE;

        (3) Drawings;

        (4) Log books;

        (5) Supervision;

        (6) References; and

        (7) Tools and test equipment.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Basic Maintainer shall perform deck and hull equipment rounds IAW refs to include:

    a. Inspecting Auxiliary boats and lifting appliances by:

        (1) Maintaining auxiliary boats and lifting appliances; and

        (2) Conducting daily/weekly rounds on auxiliary boats.

    b. Inspecting the Gasoline (F67) Fuel system;

    c. Maintaining lifting appliance;

    d. Monitoring a liquid level management system (LLMS) console.

4. Reference: A1, A2, A49, A53, A54, A55, A56, A57, A59, andA88.

5. Remarks:

   a. References at the teaching point level will need to refined during the pilot serial.

PO 006 PERFORM MAIN REFRIDGERATION & HVAC SYSTEM ROUNDS

1. Performance: Perform Main Refrigeration and HVAC System Rounds.

2. Conditions:

    a. Given:

        (1) HARRY DEWOLF class ship;

        (2) PPE;

        (3) Drawings;

        (4) Log books;

        (5) Supervision;

        (6) References; and

        (7) Tools and test equipment.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Basic Maintainer shall perform main refrigeration and HVAC rounds IAW refs to include:

    a. Inspecting refrigeration and A/C by:

        (1) Operating locally, main refrigeration.

        (2) Inspecting air conditioning and heating systems by:

            (a) Operating HVAC hot water system; and

            (b) Operating chilled water system.

4. Reference: A70, A71, A72, and A73.

5. Remarks:

a. References at the teaching point level will need to refined during the pilot serial.

PO 007 PERFORM DAMAGE CONTROL SYSTEM ROUNDS

1. Performance: Perform Damage Control System Rounds.

2. Conditions:

   a. Given:

      (1) HARRY DEWOLF class ship;

      (2) PPE;

      (3) Drawings;

      (4) Log books;

      (5) Supervision;

      (6) References; and

      (7) Tools and test equipment.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Basic Maintainer shall perform damage control system rounds IAW refs to include:

   a. Inspecting fixed fire extinguishing systems;

   b. Inspecting Seawater cooling system in emergency configuration;

   c. Inspecting Portable Firefighting Equipment;

   d. Inspecting Seawater Service system; and

   e. Inspecting Smoke Clearing HVAC system.

4. Reference: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84.

5. Remarks: Nil.

PO 008 RESPOND TO EMERGENCY CONDITIONS

1. Performance: Respond to Emergency Conditions.

2. Conditions:

    a. Given:

        (1) HARRY DEWOLF class ship;

        (2) PPE;

        (3) Drawings;

        (4) Log books;

        (5) Supervision;

        (6) References; and

        (7) Tools and test equipment.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Basic Maintainer shall respond to emergency conditions IAW refs by:

    a. Performing at sea emergency procedures by:

        (1) Operating F/O system locally;

        (2) Operating steering system locally to include:

            (a) Operating steering systems manually.

        (3) Operating OG&D systems.

    b. Rectifying alarms, to include:

        (1) Configuring power supplies.

4. Reference: A1, A2, A15, A18, A30, A29, A41, A46, A47, A72, A57, A12, A13, A37, A61, A82, and D1.

5. Remarks:

a. References at the teaching point level will need to refined during the pilot serial.

RDIMS #1230825

TRAINING PLAN

CHAPTER 3: – TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. OJT.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 90 day to complete the OJPR.

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Course Director				
Rank	MOSID	Qualifications Required	Qty	PO / EO
PO1	00379	Training PO	1	ALL
Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
PO2/MS	00124	Legacy Hull Technician – Mechanical Specialist; and PLQ	1	
PO2/MS	00367	Legacy Marine Engineer – Mechanical Specialist; and PLQ	1	
PO2/MS	00125	Legacy Marine Electrician – Electrical Specialist; and	1	

		PLQ		

Guest Lecturers/Others (May be civilian)				
Lecturer	# Instructor	Subject	EO #	# period

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

a. This course is intended to be exclusively thought via OJT which requires an operational platform and training billets.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

a. Optimum – 1;

b. Maximum Load – 1; and

c. Minimum – 1.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NTORD 4540-2).

7. The authority to load members onto this course will be IAW NTORD 4540-43. NTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. The Career Manager for career courses; or

b. NFS Campus Support for occupation and specialty courses.

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NTORD 4500-51. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. Select one of the following: There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377 and/or CF289 Certificate of Military Achievement, shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance. Additional comments must be provided on the following:

a. Training material that was not covered;

b. Areas of strengths and areas for improvement; and

c. Recommendation for future employment or training.

CHAPTER 4: LESSON SPECIFICATIONS

PO 001 – PERFORM MAIN PROPULSION SYSTEM ROUNDS

EO 001.01 INSPECTING SHAFTING AND PROPELLER SYSTEMS

1. Performance: Inspecting Shafting and Propeller Systems.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect shafting and propeller systems by:

    a. Operating shaft line turning gear locally;

    b. Operating shaft line stern tube inflatable seal:

        (1) Operating shaft locking/braking device;

    c. Inspecting the Propulsion Motors; and

    d. Operating Quality water package system locally.

    e. Inspect Main Propulsion Converter to include:

        (1) Inspecting converter cooling cabinet;

        (2) Inspecting converter control cabinet; and

        (3) Inspecting propulsion transform cooling fans.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Deliver a shipboard tour of the Shaftline Arrangement, to encompass: a. Turning gear motors; b. Local controller for turning gears; c. Sterntube inflatable seals; d. Shaft lock; e. Shaft brakes; and f. Bulkhead seals.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A7, A8, A9, A87, A90, A91, and A12. Remarks: Nil.
Operate the shaftline arrangement, to include: a. Turning gear grease routine; b. Engaging / disengaging turning gear; c. Turning gear oil type; d. Describe inflatable seal operation procedures; e. Inspect shaftline seal; f. Shaft lock local operation; and g. Shaftline brake local operation (Under supervision).	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A7, A8, A9, A87, A90, A91, and A12. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Deliver a shipboard tour of the Propulsion Motors to encompass: a. Jacking Pumps and associated instrumentation; b. Jacking Pump Controllers c. Oil sight glass, DE and NDE; d. Liquid Cooling inlet/outlet; and location of fans/controllers e. Shaft Grounding brushes.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A7, A8, A9, A87, A90, A91, and A12. Remarks: Nil.
Inspect the Propulsion motors to include: a. Oil levels; b. Cooling valve positions; and c. Local Gauges.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A7, A8, A9, A87, A90, A91, and A12. Remarks: Nil.
Deliver a shipboard tour of the Water Quality Package, to encompass: a. Pumps; b. Inlet and overboard discharge valves; c. Interfacing systems to include SW cooling system; d. Cross Connect valve and e. Local gauges.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A7, A8, A9, A87, A90, A91, and A12. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the sterntube bearing cooling system, to include: a. Flash-up / shutdown procedures; b. Seawater strainer cleaning; c. Expected local parameters; d. Blow-down procedures; and e. Rectify Air Lock.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A7, A8, A9, A87, A90, A91, and A12. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 001.02 INSPECTING MAIN PROPULSION CONVERTER

1. Performance: Inspecting Main Propulsion Converter.

2. Conditions:

   a. Given:

      (1) System schematics;

      (2) Access to systems;

      (3) Required tools;

      (4) PPE;

      (5) References; and

      (6) Supervision.

   b. Denied: Nil.

   c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect main propulsion converter by:

   a. Inspecting converter cooling cabinet;
   b. Inspecting converter control cabinet; and
   c. Inspecting propulsion transform cooling fans.

4. Teaching Points:

Teaching Points	**Instructional Strategies**	**Time**	**References/Remarks**
Inspect the Converter Cooling Cabinet System to include; a. Operate Converter HMI; and b. Associated Instrumentation.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A7, A8, A9, A87, A90, A91, and A12. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Inspect the Converter Control Cabinet to include; a. Reset Watchdog alarm; and b. Operate HMI.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A7, A8, A9, A87, A90, A91, and A12. Remarks: Nil.
Inspect the Propulsion Transformer Cooling Fans to include; a. Normal operation; and b. MCC location.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A7, A8, A9, A87, A90, A91, and A12. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

PO 002 – PERFROM PRIMARY PG&D ROUNDS

EO 002.01 INSPECT MAN DIESEL ENGINES

1. **Performance: Inspect Man Diesel Engines.**

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect MAN diesel engines IAW refs by:

    a. Identifying key components to include:

        (1) Pre-lube pump;

        (2) Local control panels;

        (3) Local gauges;

        (4) Fuel Oil Services supply & return valves;

        (5) Start air receivers;

        (6) Emergency stop buttons;

        (7) Lube oil pump;

        (8) Fresh water pumps;

        (9) Preheaters;

        (10) Wet/Dry Turbocharger Cleaning points;

        (11) Turning motor;

(12) Normal operating parameters;

(13) Trips and alarms;

(14) Air intakes;

(15) Lube oil filter;

(16) Oil mist detector;

(17) Jacking Pumps and

(18) Fuel Filters.

b. Start up/Shut down MAN diesel engine (with assistance) to include:

(1) Turbo wash procedure;

(2) Turning gear operation

(3) On watch Rounds

(4) PM routines to include:

(a) change oil;

(b) change coolant;

(c) replace intake filter;

(d) replace fuel filter;

(e) change LO filter; and

(f) record firing pressures

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components: a. Pre-lube pump; b. Local control panels; c. Local gauges; d. Fuel Oil Services supply & return valves; e. Start air receivers; f. Emergency stop buttons; g. Lube oil pump; h. Fresh water pumps; i. Preheaters; j. Wet/Dry Turbocharger Cleaning points; k. Turning motor; l. Normal operating parameters; m. Trips and alarms; n. Air intakes; o. Lube oil filter; p. Oil mist detector; q. Jacking Pumps and r. Fuel Filters	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A3, A15, A20, A22, A23, A33, A34, and A35. Remarks: Nil.
Start up/Shut down MAN diesel engine (with assistance) to include: a. Turbo wash procedure; b. Turning gear operation c. On watch Rounds	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A3, A15, A20, A22, A23, A33, A34, and A35. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
PM routines to include: a. change oil; b. change coolant; c. replace intake filter; d. replace fuel filter; e. change LO filter; and f. record firing pressures	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A3, A15, A20, A22, A23, A33, A34, and A35. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 002.01 INSPECT EXHAUST SYSTEMS

1. Performance: Inspect Exhaust System
2. Conditions:
    a. Given:
        (1) System schematics;
        (2) Access to systems;
        (3) Required tools;
        (4) PPE;
        (5) References; and
        (6) Supervision.
    b. Denied: Nil.
    c. Environment: HDW Class vessel alongside or at sea.
3. Standard: The trainee shall inspect exhaust systems, to include:
    a. Inspecting MAN Diesel Exhaust;
    b. Inspecting Incinerator Exhaust;
    c. Inspecting Water Heater Exhaust;
    d. Inspecting EHDG Exhaust:
    e. Monitor Selective Catalytic Reactor (SCR), to include:
        (1) UREA Injection Stop Mode.
4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the MAN Diesel Exhaust: a. Silencers b. SCR Unit c. Thermal Economizer and FF procedures associated with.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A3, A15, A20, A22, A23, A33, A34, and A35. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the Incinerator Exhaust: a. Incinerator; and b. Spark arrestor.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A3, A15, A20, A22, A23, A33, A34, and A35. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the Water Heater Exhaust System: a. FWD/AFT Water Heater; and b. Spark Arrestors.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A3, A15, A20, A22, A23, A33, A34, and A35. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the EHDG Exhaust System: a. EHDG; and b. Silencer.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A3, A15, A20, A22, A23, A33, A34, and A35. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the Selective Catalytic Reductions System: a. UREA storage tanks; b. Pump module; c. Dosing Unit; d. Compressed air receiver; e. Mixing Unit; f. Reactor; and g. Soot blow off system.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A3, A15, A20, A22, A23, A33, A34, and A35. Remarks: Nil.
UREA Injection Stop Mode.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A3, A15, A20, A22, A23, A33, A34, and A35. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

a. HDW vessel.

7. Remarks: Nil.

PO 003 – PERFORM MARINE ENGINEERING AUXILIARY ROUNDS

EO 003.01 INSPECT STEERING SYSTEMS

1. Performance: Inspect Steering System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect steering system IAW refs to include:

    a. Identifying key components of the steering system to include:

        (1) Normal system parameters; and

        (2) Trips and alarms.

    b. Operating the steering system locally to include:

        (1) Operating the steering system manually.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the steering system: a. Local gauges; b. Wheel pump; c. Expansion tank; d. Storage tank; e. Pumps and Local control panels; f. Circuit Breakers; g. Valve locations and configuration; and h. Alarm Panel.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Describe the theory of the Steering system, to include: a. Normal system parameters; b. Trips and alarms; and c. Oil type.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A3, A15, A20, A22, A23, A33, A34, and A35 A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83.. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the steering gear system, to include: a. Operate steering locally by motor controllers; b. Hand pump operation; c. Grease routine procedure; and d. Flash-up / shut down procedures.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.02 INSPECT BOW THRUSTER SYSTEM

1. Performance: Inspect Bow Thruster System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect the bow thruster system IAW refs by:

    a. Identifying key components of the Bow Thruster;

    b. Conducting pre-start checks; and

    c. Operating the Bow Thruster locally.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the Bow Thruster system: a. Propulsion motor; b. Coupling; c. Underwater gearbox; d. Propeller; e. Tunnel; f. Oil Expansion Tank; g. LEACON tank; h. Converter Cabinet; i. Control Cabinet; j. FW cooling Inlet/Outlet; and k. Aux MCC locations	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Conduct pre-start checks to include: a. Expansion tank oil level; b. Expansion tank valve configuration; c. LEACON tank oil level; d. LEACON tank valve configuration; e. Internal Cooling Circuit; f. HV breakers status; g. LOP power supply on; and h. Local/Remote selector switch in correct state.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the Bow Thruster locally to include: a. Selecting Local/Remote; b. Start/Stop Auxiliaries; c. Start/Stop power sequence; d. Transfer control from Local to Remote; e. Emergency stop; and f. Operate Converter Cabinet HMI.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.03 INSPECT THE FOS SYSTEM

1. Performance Inspect the FOS System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect the FOS system:

    a. Identifying key components of the FOS system; and

    b. Operating the FOS system.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the FOS system: a. Service pumps and local controllers; b. Air driven DFO supply pumps; c. Clean leakage fuel tanks; d. Clean DFO pumps; e. Manual / Automatic Filters; f. Supply / return mechanically interlocked valves; g. Electric supply valves; h. Bulkhead isolations; i. Day Tanks; j. Local pressure gauges; k. Pressure regulators; l. Quick closing valves; m. Flow Meters; n. Coolers; o. Maintenance valve; p. Pressure retaining valve; and q. Oil mist lubricator	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Describe the theory of the FOS, system onboard AOPV, to include: a. Normal System parameters; b. Emergency / Abnormal operations; and c. Automatic functions.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the fuel oil service system, to include: a. Manually open day tank suction and discharge valves; b. Valve charging status (batteries); c. Service pump flash-up / shut down procedures; d. Operate quick closing valves; e. Air driven DFO pump flash-up / shut down procedures; f. Clean DFO pump flash-up / shut down procedures; and g. Automatic Fuel Filter.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

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7. Remarks: Nil.

EO 003.04 INSPECT THE FOT SYSTEM

1. Performance: Inspect the FOT System.
2. Conditions:
    a. Given:
        (1) System schematics;
        (2) Access to systems;
        (3) Required tools;
        (4) PPE;
        (5) References; and
        (6) Supervision.
    b. Denied: Nil.
    c. Environment: HDW Class vessel alongside or at sea.
3. Standard: The trainee shall inspect the FOT system:
    a. Identifying key components of the FOT system; and
    b. Operating the FOT system.
4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the FOT system: a. Fwd / Aft fuel manifolds; b. Bulkhead isolations; c. Settling tank valve locations; d. DFO Transfer Pumps; e. RAS stations / bunkering stations. f. DFO Boat and EHDG Transfer pump; g. DFO Boat Fueling Stations; h. DFO Tank vent locations; i. Quick Close Valve locations/Reset procedure j. Overflow tank / vent; and k. Sounding tube locations.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Describe the theory of the FOS, system onboard AOPV, to include: a. Normal System parameters; b. Emergency / Abnormal operations; and c. Automatic functions	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83.. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the FOT system, to include: a. Flash up / shut down procedures of the Fuel Transfer Pumps; b. Fuel manifold configuration; c. Valve charging status (batteries); d. Flash-up / shut down procedures of the Boat and EHDG Transfer pump; and e. Conduct rounds on FOT system.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.05 INSPECT FUEL/LUBE OIL PURIFIERS

1. Performance: Inspect Fuel/Lube Oil Purifiers.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect fuel/lube oil purifiers, to include:

    a. Identify key components of the Fuel oil purifiers;

    b. Operate Fuel oil purifiers;

    c. Identify key components of the Lube;

    d. Operate Lube oil purifiers.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the Fuel oil purifiers: a. Settling Tanks; b. DFO Purifiers; c. DFO Purifier Feed Pumps; d. DFO Purifier Heaters; and e. Associated valves and piping.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the Fuel oil system, to include: a. Flash-up / shutdown of Purifier; and b. Perform De-sludge Cycle Manually; and c. Navigate the HMI.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the Lube oil purifiers: a. Local control panels; b. L.O. Purifier supply pump; c. L.O. Purifier pre-heaters; and d. L.O. Purifier interfacing connections / valves.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the Lube Oil Purifier system to include: a. Start / Stop L.O. Purifier pre-heaters; b. Start / Stop L.O. Purifiers; c. Define expected operating parameters; and d. Explain emergency shutdown procedures.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.06 INSPECT DIESEL LUBE OIL FILLING AND TRANSFER SYSTEM

1. Performance: Inspect Diesel Lube Oil Filling and Transfer System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect diesel lube oil filling and transfer system to include:

    a. Identifying key components of the diesel lube oil system;

    b. Operating diesel lube oil system;

    c. Identifying key components of the diesel lube oil purifier system; and

    d. Operate the diesel lube oil purifier system.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the diesel lube oil system: a. Port and STBD Filling Stations; b. Pressure regulators; c. Oil Traps; d. LO Tanks; e. LO Pumps; f. Waste Oil Tank; g. Waste Oil Pump; h. Local gauge panels; i. Tank sounding locations; j. Bunkering valve locations; and k. DG filling valves.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the diesel lube oil system, to include: a. Flash-up / shut down procedures of LO Transfer Pumps; b. Transfer oil between LO tanks and DG's; c. Transfer Oil from DG's to Waste Oil Tank; and d. Transfer waste oil from waste oil tank to sludge tank.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the diesel lube oil purifier system: a. Local control panels; b. L.O. Purifier supply pump; c. L.O. Purifier pre-heaters; and d. L.O. Purifier interfacing connections / valves.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the L.O. Purifier system to include: a. Start / Stop L.O. Purifier pre-heaters; b. Start / Stop L.O. Purifiers; c. Define expected operating parameters; and d. Explain emergency shutdown procedures.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.07 INSPECT COMPRESSED AIR SYSTEMS

1. Performance: Inspect Compressed Air Systems.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect compressed air systems, to include:

    a. Inspecting Ship's Service Air by:

        (1) Verifying the operation of the Ship's Service (SS), Ship's Start (SST) and Breathing Air systems;

        (2) Identifying key components of the SS Air system; and

        (3) Operating the SS Air system locally.

    b. Inspecting Ship's Start Air by:

        (1) Identifying key components of the SST Air system; and

        (2) Operating the SST Air system locally.

    c. Inspecting Breathing Air Compressors by:

        (1) Identifying key components of the Breathing Air Compressors; and

        (2) Operating the Breathing Air Compressor locally.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Verify the operation of the SS, ST and Breathing Air systems found onboard an AOPV, to include: a. Normal system pressure; and b. Oil and coolant type.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the SS Air System: a. Compressors and local controllers; b. SS air dryers; c. Dryer in/out & bypass valves; d. Flask inlet valves; e. End users; f. Flask drain valves; and g. Regulators.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate SS Air System to include: a. Flash-up / Shut down of SS air compressor; b. Alarms and trips; c. Rounds on Dryer / Compressor; and d. Reducing stations.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the SST Air system: a. Compressors and local controllers; b. Flask inlet valves; c. Flask drain valves; d. Regulators; and e. SST to SS Air cross connect.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the SST Air system: a. Flash-up / Shut down of SST air compressor; b. Alarms and trips; c. Rounds on compressor; d. Procedure for SST to SS Air cross connect.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the Breathing air compressors: a. Breathing Air filling station; b. Diving Air Compressor; and c. Portable Breathing Air compressor.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the breathing air compressor to include: a. Procedure for filling bottles from both the booster station and the portable breathing air compressor.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.08 INSPECT FRESH WATER SYSTEM

1. Performance: Inspect Fresh Water System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect fresh water system, to include:

    a. Inspecting the cold fresh water system, to include:

        (1) Operating the Fresh Water Services system locally.

    b. Inspecting the Hot Fresh Water system, to include:

        (1) Operating locally the Hot Fresh Water system.

    c. Inspecting the FW Purification system, to include:

        (1) Operating the FW Purification system.

    d. Inspecting the Reverse Osmosis Desalination (ROD) system, to include:

        (1) Operating the ROD locally.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Describe the basic operation of, normal system pressure and alarm points for fresh water systems onboard an AOPV, to include: a. fresh water; b. hot water; c. Chlorine dosing; d. Technical Water, and e. ROD Systems.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the fresh water system, to include: a. Upper deck filling valves; b. Upper deck riser isolation valves; c. Tank filling valves; d. Transfer main cross connect to F/W main; e. Local Pressure gauges; f. Pumps and local control panel; g. Pump supply & discharge valves; h. Leak off valves; i. UV Sterilizer; j. Ion Exchange Filter; and k. Supply and return main isolation valves.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate fresh water system, to include: a. Hydrophore tanks; b. Procedure for starting/stopping F/W pumps; c. Procedure for starting/stopping Hot Water Calorifiers; and d. Procedure for TEU support systems set up.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the hot fresh water, to include: a. Freshwater Calorifiers; b. Domestic H/W recirculation pumps; c. H/W expansion tank; d. Riser isolation valves to HFW main; e. Upper deck H/W bibs / Helicopter wash down bib; f. Cold fresh water supply valves; and g. Location of relief valves.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the hot fresh water, to include: a. Local and Remote start/stop of the calorifiers and pumps.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the freshwater purification system, to include: a. Location of equipment; b. Chlorine dosing system; and c. UV Sterilizer.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the freshwater purification system (chlorine dosing system), to include: a. Configure the system to treat either tank; b. Configure inline dosing operation; c. Operate the pump; and d. Procedure for treating a F/W Tank	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the RODs, to include: a. Location of valves; and b. Location of filters.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the ROD system, to include: a. Start/Stop Procedures; b. Preserving; c. Filter changes; and d. Conduct rounds.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.09 INSPECT BLACK AND GREY WATER COLLECTION/TREATMENT SYSTEMS

1. Performance: Inspecting Black and Grey Water Collection/Treatment System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect black and grey water collection/treatment systems IAW refs, to include:

    a. Inspecting the Black Water (BW) system by:

        (1) Operating the BW system.

    b. Inspecting the Grey Water (GW) system by:

        (1) Configuring the GW system.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Describe the basic operation of, normal system pressure and alarm points for black and water systems onboard an AOPV.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the water system, to include: a. Local control panel; b. Raw sewage discharge pumps; c. Treated sewage discharge pumps; d. Macerators; e. Upper deck discharge; f. Flushing connection; g. Cross connect between GW and BW, h. Treated sewage discharge pumps supply / discharge and overboard valves; i. Discharge pumps; and j. Sludge blow down.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the black water, to include: a. Treated black water system operation; b. Non-treated black water system operation; c. Procedure for discharging ashore/at sea; and d. Bio Reactor.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the Grey water system, to include: a. Grey water transfer Pp; b. Gray Water Interface Units c. Black water transfer Pp; d. Black/grey water vacuum pumps; and e. Grease separator.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Configure the Grey water system, to include: a. Inboard use; and b. Outbaord use.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.10 INSPECT THE OIL POLUTION AND ABATEMENT SYSTEM

1. Performance: Inspect the Oil Pollution and Abatement (OPA) System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect the oil pollution and abatement system IAW refs, to include:

    a. Conducting rounds on the oil pollution and abatement system; and

    b. Operating the OPA.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Conducting rounds on the OPA system, to include: a. Local Control Panels; b. Pumps; c. Suction / Discharge valve locations (inside and outside machinery spaces); d. Bulkhead isolation valves; e. Filters; f. Oily Water Separator and associated valves; g. Mobile Separator and associated valves; h. Tank locations; and i. Tank sounding locations	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the OPA system, to include: a. Valve configuration for pumping to upper decks; b. Valve configuration for all interfacing OPA sub-systems; c. Start / Stop all OPA pumps; d. Start / Stop Oily Water Separator; e. Maintain Oil Content Monitor; f. Configure valves to discharge clean water overboard / back to storage tanks; g. Start / Stop Mobile Separator; h. Configure Mobile Separator suction / discharge valves; and i. Define expected operating parameters. j. Valve configuration from Oily water to sludge system k. Valve configuration for filling incinerator sludge tank.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

RDIMS #1230825

EO 003.11 INSPECT BALLAST WATER TREATMENT SYSTEM

1. Performance: Inspect Ballast Water Treatment System (BWTS).

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect ballast water treatment system IAW refs, to include:

    a. Conducting pre-start checks; and

    b. Operating BWTS.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Describe the operation of the BWTS to include: a. Normal system operation; b. Manual Backflushing operation; c. Automatic Backflushing; d. BWTS bypass operation; and e. Abnormal operation.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the water system, to include: a. Master control panel; b. By-pass valve; c. Differential Pressure Transmitter; d. Pressure Transmitter; e. Flowmeter; f. Sample point/sample valve; g. UV Unit; h. Pumps; and i. Automatic Filter.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the BWTS to include: a. Prestart Checks; b. By-pass configuration; c. Navigate HMI; d. Conduct rounds; and e. Emergency Shutdown Procedures.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.12 INSPECT BALLAST WATER SYSTEM

1. Performance: Inspect Ballast Water System.

2. Conditions:

   a. Given:

      (1) System schematics;

      (2) Access to systems;

      (3) Required tools;

      (4) PPE;

      (5) References; and

      (6) Supervision.

   b. Denied: Nil.

   c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect ballast water system IAW refs, to include:

   a. Operating the ballast water system.

4. Teaching Points:

Teaching Points	**Instructional Strategies**	**Time**	**References/Remarks**
Describe the operation of the BWS to include: a. Normal system operation; b. Abnormal System Operation; and c. Normal system pressures.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the ballast water system, to include: a. Tank location; b. Priming pumps and air supplies; c. FWD and AFT supply and return valves; d. Tank valves; e. Bilge/Ballast X-Conn Valves; f. Pump Controllers; g. Pump breakers; and h. Bilge/Ballast Pumps.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the BWS to include: a. Manual operation of valves; and b. Locally Start/Stop pumps.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

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EO 003.13 INSPECT THE INCINERATOR SYSTEM

1. Performance: Inspect the Incinerator System.

2. Conditions:

   a. Given:

      (1) System schematics;

      (2) Access to systems;

      (3) Required tools;

      (4) PPE;

      (5) References; and

      (6) Supervision.

   b. Denied: Nil.

   c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect the incinerator system IAW refs, to include:

   a. Operating the incinerator system.

4. Teaching Points:

Teaching Points	**Instructional Strategies**	**Time**	**References/Remarks**
Describe the operation of Incinerator system to include: a. Solid Operation; b. Solid/Sludge Operations.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the incinerator system, to include: a. Incinerator unit; b. Local control panels; c. Incinerator fuel tank; d. Mixing Tank Unit; e. Feed pump; f. Mixing pump; g. Transfer pump; h. Air nozzles; i. Sludge oil filter; j. Sludge oil dosing pump; k. Fire eyes; l. Associated HVAC Control Panel; m. Emergency Stop; n. Quick Closing Valve; o. Blower units; and p. Breaker panel.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the Incinerator system to include: a. Flash-up / shutdown incinerator in solid mode; b. Fuel tank filling operations (DFO/Sludge); c. Flash-up / shutdown incinerator in solid/sludge mode; d. HVAC Breaker reset procedure; e. Operating procedures post blackout; and f. Configure valves for operation.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.14 INSPECT EQUIPMENT COOLING SYSTEMS

1. Performance: Inspect Equipment Cooling Systems.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied: Nil.

    c. Environment: HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect equipment cooling systems IAW refs, to include:

    a. Inspecting the Central Fresh Water Cooling (CFWC) system to include:

        (1) Operating the CFWC system.

    b. Inspecting the Seawater Cooling (SWC) system to include:

        (1) Operating the SWC system.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Describe the operation of the CFWC system onboard an AOPV, to include: a. Normal system configuration; b. IPMS control; c. Pump and cross connect auto functions; and d. System pressures.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the central FW cooling system, to include: a. Pumps and controllers; b. Pump suction & discharges; c. Heat exchanger isolation valves; d. Chillers; e. Refrigeration system three-way valves; f. DFO coolers; g. Strainers; h. Local gauge panels; i. Auxiliary Cooling Water System; j. Dump Cooler Cooling Water System; k. EHDG Cooling Water System; l. maintenance water filling pump; and m. f/w fwd/aft cross connects.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the CFWC system, to include: a. Local/Remote operation of pumps; b. Conduct OCCAP; c. Cross connect between Cold Fresh Water System and CFWC system; d. Valve alignment for filling MDG and EHDG; and e. Take/analyze samples.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Describe the operation of the SWC system onboard an AOPV, to include: a. Normal system configuration; b. Cold weather procedures c. IPMS control; d. Pump and cross connect auto functions; and e. System pressures.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the central FW cooling system, to include a. Pumps and controllers; b. Pump suction & discharges; c. Heat exchanger isolation valves; d. Sea Chest; e. Sea Bay's; f. Strainers; g. Local gauge panels; h. Re-circulation and overboard valves; i. MCC locations; and j. Sea bay vents.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the SWC system, to include: a. Local/Remote operation of pumps; b. Normal operation; c. Alongside operation; d. Emergency operation for Bilge suction; and e. Valve configurations.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

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7. Remarks: Nil.

EO 003.15 INSPECT THE THERMAL OIL SYSTEM

1. Performance: Inspect the Thermal Oil System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect the thermal oil system IAW refs, to include:

    a. Verifying basic operating parameters of the thermal oil system;

    b. Operating the thermal oil system locally.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Verify basic operating parameters of the Thermal Oil system to include: a. Normal operation; b. Emergency operation; and c. Normal system parameters.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the thermal oil sytem, to include: a. Local control panels; b. Circulation pumps; c. Economizers; d. Dump coolers; e. Exhaust Heat Recovery Unit (EHR); f. De-aerator tank; g. Expansion tank; h. Emergency drain tank; i. Header tank dump valve; j. Economizer by-pass valves; k. Header tank filling pump / associated valves; l. Economizer 3-way valve change over switches; m. f/w cleaning nozzle valves; n. damper actuators; o. emergency FF sea water valves; and e. Manual change over switch (ELEC).	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the Thermal oil system locally, to include: a. Flash-up/Shutdown procedures; b. Monitor local gauges; c. Change over switches (ELEC); d. Perform OCCAP samples; and e. Emergency operations: a. Thermal oil shut off valves; and b. Emergency stops.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.16 INSPECT THE TANK HEATING SYSTEM

1. Performance: Inspect the Tank Heating System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect the tank heating system IAW refs, to include:

    a. Verfying basic operating parameters;

    b. Operating the tank heating system.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Verify basic operating parameters of the Tank Heating System to include: a. normal operating parameters: b. normal system operation.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the tank heating sytem, to include: a. Circulation pumps; b. Heat Exchanger; c. Tank locations; d. Tank valves; e. Contamination Monitor; f. Expansion tank; g. Breaker locations for pumps; and h. Remote-operated control valves.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the Tank Heating system to include: a. Operate tank valves in manual; b. Preform sample analysis test; c. Start/Stop pumps from local panels; and d. Perform OCCAP sample.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 003.17 INSPECT STABILITY SYSTEMS

1. Performance: Inspect Stability Systems.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect stability systems IAW refs, to include:

    a. Inspecting Active Roll Fin Stabilization system to include:

        (1) Operating the Active Fin Roll Stabilization system locally.

    b. Inspecting Heeling system to include:

        (1) Operate the Heeling system locally.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Describe the operation of the Fin Stabilizer system: a. Normal system operation; and b. Oil and coolant type.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the Fin Stabilizer sytem, to include: a. Local controller; b. Fin unit Assembly; c. Main hydraulic pump; d. Supply tank; e. Oil Cooler / Heat exchanger; f. Emergency Pump; g. Instrumentation and gauges; and h. Lube oil tank.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate the Fin Stabilizer system to include: a. Prestart Checks; b. Alarms and trips; c. Manual operation and emergency stop; d. Rounds on hydraulic systems; and e. Emergency fin housing.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Describe the operation of the Heeling system to include: a. Normal operation; and b. Manual operation.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the Heeling system to include, to encompass: a. Control System; b. Heeling Tanks; c. Heeling Pumps; and d. Associated valves.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.
Operate Heeling system to include: a. Flash-up / shut down procedures; b. Valve configurations; c. Manual operation; and d. Emergency stop.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A61, A62, A63, A12, A13, A37, A10, A11, A14, A46, A47, A52, A53, A54, A67, A64, A58, A65, A59, A48, A49, A60, A39, A40, A44, A45, A46, and A83. Remarks: Nil.

5. Total EO Time as follows:

a. Conducted during 3 month OJT.

6. Training and Learning Aids:

a. HDW vessel.

7. Remarks: Nil.

PO 004 – PERFROM SECONDARY PG&D ROUNDS.

EO 004.01 Inspect Secondary PG&D.

1. Performance: Inspect Secondary PG&D.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect secondary PG&D IAW refs, to include:

    a. Inspecting Secondary PG&D systems to include;

        (1) Operating LV power generation systems remotely; and

        (2) Operating 440V distribution system.

    b. Inspecting electrical LV switchboard remotely to include:

        (1) Operating LV and Emergency SWBD locally;

        (2) Paralleling electrical loads remotely; and

        (3) Operating switchboard breakers remotely.

    c. Testing UPS to include:

        (1) Inspecting uninterruptable power supplies wiring harness.

        (2) Testing switching unit auto to include;

(3) Testing switching unit manual..

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Observing safety precautions to include: a. PPE; b. Connection procedures; and c. Verify calibration of equipment.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.
Describe the operation and status of the secondary PG&D systems to include: a. Main Diesel generators; b. PORT and STBD LV Switchboards (alarm conditions, status of power generation, checking grounds, and emergency procedures); c. Emer. SWBD; d. Emergency harbor Diesel; e. Distribution transformers; f. Circuit breakers and switch gear (including Breaker identification); and g. UPS operation (including start procedures, faults and locations).	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks

Identify the location of, basic functionality and relevant safety considerations for the following key components of the secondary PG&D systems to include:
 	
a. Diesel generators;
b. The PORT and STB LV Switchboards;
c. Port and STBD 120V switchboards
d. 220V Distribution;
  a. De-Icing; and
  b. Distribution panels and associated equipment.
e. Circuit Breaker Locations;
f. MCC locations;
g. 440V distribution panels;
h. Fitted UPS locations; and
i. The Static frequency converter.
	Method: OJT
	
	Media: Real Equipment
	
	Environment: HDW vessel
		Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42.
			
			Remarks: Nil.

Inspect LV switchboard to include: a. Operate LV and Emergency Switchboard locally; b. Paralleling electrical loads locally; and c. Operate Switchboard breakers locally.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Testing UPS, to include: a. Associated alarms; and b. Inspect Uninterruptable Power Supply wiring harness.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 004.02 INSPECT ELECTRICAL SYSTEMS

1. Performance: Inspect Electrical Systems.

2. Conditions:

   a. Given:

      (1) System schematics;

      (2) Access to systems;

      (3) Required tools;

      (4) PPE;

      (5) References; and

      (6) Supervision.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect electrical systems IAW refs, to include:

   a. Inspecting the navigational system; and

   b. Testing shipboard alarms.

4. Teaching Points:

Teaching Points	**Instructional Strategies**	**Time**	**References/Remarks**
Identifying the location of: a. The navigational, search, and signal lighting; b. The Whistle/foghorn; and c. The feeder circuits.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Inspect the navigational systems to include: a. Navigational lights; b. Ships whistles; c. Timing circuit; and d. Navigation light inter connection.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.
Testing shipboard alarms: a. Dead man alarm system; b. Noxious gas detection system; c. Crash on deck; and d. Locked in fridge alarm.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 004.03 INSPECT IMPRESSED CURRENT CATHODIC PROTECT SYSTEM (ICCP)

1. Performance: Inspect Impressed Current Cathodic Protect System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect ICCP system IAW refs, to include:

    a. Operating the ICCP system.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the ICCP systems to include: a. The remote controller; b. The anodes; c. The reference electrodes; d. Shaft grounding slip rings; e. current meters; f. Shaft grounding Amp meters; and g. The feeder circuit locations.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.
Operate the ICCP system to include: a. Energizing power feeds; b. Energizing ICCP mains; c. Cleaning particulate from slip rings; d. Observe fault indicator condition; and e. Portable reference electrode.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

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EO 004.05 INSPECT SMALL BOAT ELECTRICS

1. Performance: Inspect Small Boat Electrics.

2. Conditions:

   a. Given:

      (1) System schematics;

      (2) Access to systems;

      (3) Required tools;

      (4) PPE;

      (5) References; and

      (6) Supervision.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect small boat electronics IAW refs, to include:

   a. Applying PPE protocols.

4. Teaching Points:

Teaching Points	**Instructional Strategies**	**Time**	**References/Remarks**
Applying PPE protocols to include: a. Use of personal floatation device; b. MOBI; and c. RADHAZ safety.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Inspecting small boat electrics to include a. Display console lights and displays; b. Diesel engine connections; c. Batteries connections and cleanliness; d. Navigational lights; e. Searchlights; f. Horn; g. Bilge pumps; and h. Charging circuit.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 004.06 OPERATE DECK MACHINERY

1. Performance: Operate Deck Machinery.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall operate deck machinery IAW refs, to include:

    a. Operating the winch power units in all modes; and

    b. Operating the capstans and anchor windlass.

4. Teaching Points:

Teaching Points	**Instructional Strategies**	**Time**	**References/Remarks**
Operate winch power units to include: a. Identifying breaker locations; b. Reacting to overload trips; c. Start/stops; and d. Operating speed controls.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operating capstans and windlass to include: a. Identifying breaker locations; b. Reacting to overload trips; c. Start/stops; and d. Operating speed controls.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 004.07 CONDUCT ELECTRICAL PRE-SAIL CHECKS

1. Performance: Conduct Electrical Pre-Sail Checks.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall conduct electrical pre-sail checks IAW refs, to include:

    a. Steering pre-sail agreement checks.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Conduct Pre-sail Checklist, to inlcude: a. Obtain and conduct pre-sail checklist as per SSO's b. Conduct steering pre-sail c. . Lamp test all steering positions d. Ensure system is free of alarms and warnings e. All feeders are energized f. Aid in steering checks as per SSOs.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 004.08 INSPECT THE EMERGENCY HARBOUR DIESEL GENERATOR (EHDG)

1. Performance: Inspect the Emergency Harbour Diesel Generator (EHDG).

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect the EHDG IAW refs, to include:

    a. Operating the EHDG.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the EHDG, to include: a. Expansion tank; b. Speed governor; c. Pre-lube pump; d. Batteries; e. Local control panel; f. Local gauges; g. Fuel Tank supply & return valves; h. Emergency stop button; i. Lube oil pump; j. Fresh water pump; k. Preheaters; l. Turning motor; m. Normal operating parameters; n. Trips and alarms; o. Air intakes; p. Hvac and controller resets; q. Fuel filters; r. FOT filling valves; s. Oil cooler; and t. Dip stick.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the EHDG, to include: a. Cold start procedures; b. Flash-up / shut down procedures; c. Reset from emergency shut down; d. Oil and coolant type and filling/draining procedures; and e. Conduct rounds on diesel generator.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A2, A4, A4, A15, A16, A17, A18, A22, A25, A28, A29, A30, and A42. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

PO 005 – PERFORM DECK AND HULL EQUIPMENT ROUNDSANCE STATEMENT

EO 005.01 INSPECT AUXILIARY BOATS

1. Performance: Inspect Auxiliary Boats.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect auxiliary boats IAW refs, to include:

    a. Conducting rounds on Lifeboats;

    b. Conducting rounds on MRRB’s;

    c. Conducting rounds on RHIB’s; and

    d. Conducting rounds on Landing Craft.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Perform rounds on the Lifeboats, to include checks on: a. Lubricating oil; b. Cooling water system; c. Fuel system; d. Propeller shaft arrangement; and e. Starting of the engine.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, and A2. Remarks: Nil.
Perform rounds on the MRRB's to include checks on: a. Bilge level; b. Battery charge; c. Oil level; d. Cooling systems; e. Horn; f. Navigation System; g. Belt tension; h. Extinguisher system; i. Pontoon inflation; j. Navigational lights; and k. Pre-heating system.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, and A2.. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Perform rounds on the RHIB's to include checks on: a. Bilge level; b. Battery charge; c. Oil level; d. Cooling systems; e. Horn; f. Navigation System; g. Belt tension; h. Extinguisher system; i. Pontoon inflation; j. Navigational lights; k. Charging cable; and l. Breaker for charging system.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, and A2. Remarks: Nil.
Perform rounds on the Landing Craft to include checks on: a. Bilge level; b. Battery charge; c. Oil level; d. Cooling systems; e. Horn; f. Navigation System; g. Belt tension; h. Extinguisher system; i. Navigational lights; and j. Charging cable.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, and A2. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 005.02 INSPECT THE GASOLINE (F67) FUEL SYSTEM

1. Performance: Inspect the Gasoline (F67) Fuel System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect the gasoline (F67) fuel system IAW refs, to include:

    a. Maintaining the gasoline (F67) fuel system.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the gasoline (F67) system, to include: a. Drop tank locations; b. Gasoline storage locations; c. Auxiliary boat fueling locations; d. Associated valves and instrumentations; e. Control cabinet; and f. Hose connections.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A88. Remarks: Nil.
Maintain Auxiliary fueling locations by performing: a. Grease routines; b. Heat trace inspections; and c. Connections and fittings.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A88. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 005.03 INSPECT LIFTING APPLIANCES

1. Performance: Inspect Lifting Appliances.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inpsect lifting appliances IAW refs, to include:

    a. Conducting rounds on davits.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the lifting appliance, to include: a. Location of Davits; b. Control Panels; c. Motors; and d. Arm/winch.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, and A2. Remarks: Nil.
Conduct rounds on davits, to include: a. Operational checks.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, and A2. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 005.04 INSPECT LIQUID HANDLING SYSTEMS

1. Performance: Inspect Liquid Handling Systems.

2. Conditions:

   a. Given:

      (1) System schematics;

      (2) Access to systems;

      (3) Required tools;

      (4) PPE;

      (5) References; and

      (6) Supervision.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inpsect liquid handling systems IAW refs, to include:

   a. Monitoring liquid handling systems locally.

4. Teaching Points:

Teaching Points	**Instructional Strategies**	**Time**	**References/Remarks**
Describe the theory of Hull fitted tanks, to include: a. Utilizing ships drawings to identify all of the hull fitted tanks.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A49, A53, A54, A55, A56, A57, and A59. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the liquid handling systems, to include: a. Ballast; b. Potable water; c. Grey water; d. Black water; e. Void; f. DFO; g. JP5; h. UREA; i. Heeling tanks; j. Recovered oil; k. Oily Water; l. Maintenance water; m. Thermal Oil; n. Lube Oil; o. Waste Oil; p. Leakage Oil; q. Clean Leakage Oil; r. Sludge; s. Bilge; and t. Oil sump storage.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A49, A53, A54, A55, A56, A57, and A59. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Local level monitoring of sounding tubes / level indicators / LLM alarm panels and their locations to include: a. Ballast; b. Potable water; c. Grey water; d. Black water; e. Void; f. DFO; g. JP5; h. UREA; i. Heeling tanks; j. Recovered oil; k. Oily Water; l. Maintenance water; m. Thermal Oil; n. Lube Oil; o. Waste Oil; p. Leakage Oil; q. Clean Leakage Oil; r. Sludge; s. Bilge; and t. Oil sump storage.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A49, A53, A54, A55, A56, A57, and A59. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

PO 006 – PERFORM MAIN REFRIDERATION AND HVAC SYSTEMS ROUNDS

EO 006.01 INSPECT THE MAIN REFRIDERATION SYSTEM

1. Performance: Inspect the Main Refrideration System.

2. Conditions:

   a. Given:

      (1) System schematics;

      (2) Access to systems;

      (3) Required tools;

      (4) PPE;

      (5) References; and

      (6) Supervision.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect the main refrigeration system IAW refs to include:

   a. Operating the main refrigeration system.

4. Teaching Points:

Teaching Points	**Instructional Strategies**	**Time**	**References/Remarks**
Describe the theory of operation to include: a. Normal operation; b. System parameters; c. Trips/Alarms; and d. Oil type.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A70, A71, A72, and A73. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the liquid handling systems, to include: a. Compressor; b. Condenser; c. Evaporator; d. Thermal control valve; e. Local Gauges; f. Control panel; g. Oil sump; h. Rounds on refrigeration system; i. Drain pot; and j. Fresh Water Cooling system supply and return valve.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A70, A71, A72, and A73. Remarks: Nil.
Operate Main Refrigeration System to include: a. Start / Stop Refrigeration Compressors; b. Condenser maintenance/cleaning; and c. HMI navigation.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A70, A71, A72, and A73. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

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7. Remarks: Nil.

EO 006.02 INSPECT AIR CONDITIONING AND HEATING SYSTEMS

1. Performance: Inspect Air Conditioning and Heating System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect air conditioning and heating systems IAW refs, to include:

    a. Inspecting a Chiller to include:

        (1) Operating a chiller.

    b. Inspecting the Chilled Water system to include:

        (1) Operating the Chilled Water systems.

    c. Inspecting the Heating Ventilation and Air Conditioning (HVAC) system to include:

        (1) Operating the HVAC system.

    d. Inspecting the Hot Water HVAC system to include:

        (1) Operating the Hot Water HVAC system.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the chiller, to include: a. Chiller unit and local controllers; b. Cooling water system interfaces; c. Local gauges; d. Evaporator; and e. Condenser.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A70, A71, A72, and A73. Remarks: Nil.
Operate a Chiller to include: a. Flash-up / Shut down procedure; b. Operating parameters; c. Condenser cleaning and maintenance; d. Trips and alarms; and e. Leak testing.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A70, A71, A72, and A73. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the chilled water system, to include: a. Circulation pumps; b. Filter locations; c. Expansion tanks; d. Local gauges; e. Main isolation valves; f. Venting locations; and g. Filling locations.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A70, A71, A72, and A73. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the Chilled Water and Hydronic systems, to include: a. Prestart /shutdown; b. Rounds; and c. Charging system.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A70, A71, A72, and A73. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the HVAC system, to include: a. Smoke Dampers; b. Shut-off Dampers; c. Exhaust Dampers; and d. Local Start / Stop Panels.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A70, A71, A72, and A73. Remarks: Nil.
Operate the Heating Ventilation and Air Conditioning Systems (HVAC), to include: a. Procedure for local/remote operation of A/C Units.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A70, A71, A72, and A73. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the HVAC Hot water system, to include: a. Main isolations; b. Start / Stop Panels; and c. Local / Remote Control locations	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A70, A71, A72, and A73. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the Hot Water HVAC System to include: a. Procedure for local/remote operation of Hot Water System; and b. Procedure for changing between winter and summer settings.			Refs: A70, A71, A72, and A73. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

PO 007 – PERFORM DAMAGE CONROL SYSTEM ROUNDS

EO 007.01 INSPECT FIXED FIRE EXTINGUISHING SYSTEMS

1. Performance: Inspect Fixed Fire Extinguishing Systems.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect fixed fire extinguisher systems IAW refs, to include:

    a. Inspect the Aqueous Film Forming Foam (AFFF) system to include:

        (1) Verifying basic operating parameters of the AFFF system; and

        (2) Operating the Aqueous Film Forming Foam (AFFF) system.

    b. Inspect the Fixed Gas system to include:

        (1) Verifying basic operating parameters of the fixed gas system; and

        (2) Operating the Fixed Gas system.

    c. Inspect the Water Mist firefighting system to include:

        (1) Verifying basic operating parameters of the water fist firefighting system; and

        (2) Operating the Water Mist system.

d. Inspect the Twin Agent Unit (TAU) firefighting system to include:

(1) Verifying basic operating parameters of the TAU firefighting system; and

(2) Operating the TAU system.

e. Inspect the galley fitted fire suppression system firefighting system to include:

(1) Verifying basic operating parameters of the APC firefighting system; and

(2) Operating the APC system.

f. Inspect the Drencher firefighting system to include:

(1) Verifying basic operating parameters of the Drencher firefighting system; and

(2) Operating the Drencher system.

g. Operate watertight Sliding Door system

4. Teaching Points:

Teaching Points	**Instructional Strategies**	**Time**	**References/Remarks**
Verify basic operating parameters of the AFFF system, to include: a. Direction of flow; b. Operating pressure; c. Mix Ratio; d. Flow rates/Flow duration; e. Tank capacities; f. Level sensors; and g. Fill locations.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the AFFF system, to include: a. Location of remote activation; b. Location of isolation Valves; c. Arrangement of piping; d. Location of foam monitors; e. Location of high expansion foam proportioner; and f. Direction of flow.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Operate the AFFF system, to include: a. Activating the system locally / remotely; b. Flushing / draining system; c. Hazmat consideration; and d. Testing/PM routine.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Verify basic operating parameters of the fixed gas system, to include: a. Fixed Gas considerations and health/safety; b. Direction of flow; c. Operating pressure; d. Types of cylinders; e. Soak times; and f. Cylinder capacities.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the fixed gas system, to include: a. Location of all Fixed Gas cylinders; b. Arrangement of piping; c. Location of all spray heads; d. Location of all temperature detectors/beacons; e. Direction of flow; and f. Testing/PM routine.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Operate the fixed gas system, to include: a. Location of activation methods to include; b. Remote activation (IPMS); c. Local activation; d. Automatic activation; and e. Bottle activation.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Verify basic operating parameters of the water mist fire fighting system to include: a. Direction of flow; b. Operating pressure; c. Sea water operation; and d. Freshwater operation.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the water mist firefighting system, to include: a. Location of all actuation/manual call points; b. Control panels and release logic unit; c. Repeater panels location/operation; d. Arrangement of piping; e. Location of all isolation valves; f. Location and operation of all spray heads; g. Location of pumps; h. Location of solenoid valves; and i. Direction of flow.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Operate the water mist firefighting system, to include: a. Post activation procedures; b. Fresh water flushing procedures; c. Normal operation; d. Emergency operation; and e. Testing/PM routines.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Verify basic operating parameters of the Twin Agent Unit fire fighting system, to include: a. Direction of flow; b. Operating pressure; c. Tank/cylinder capacities/contents; d. TAU considerations; e. Health / safety; and f. Emergency operation.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the water mist fire fighting system, to include: a. Location/layout of TAU Compartment; b. Location of all activation points local/remote; c. Location of Hose reels; d. Arrangement of piping; e. Location of pressure relief valves; and f. Direction of flow.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Operate the TAU from the three locations: a. Normal Operation / Manual Operation; b. Post activation considerations; and c. Testing/PM routine.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Verify basic operating parameters of the galley fitted fire suppression system, to include: a. Direction of flow; b. Operating pressure; c. Cylinder capacity / contents; and d. APC considerations and health/safety.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the galley fitted fire suppression system, to include: a. Location of APC cylinders; b. Location of APC Extinguisher; c. Location of all activation points; d. Location of all fusible links; e. Arrangement of piping; f. Location of all spray heads; g. Direction of flow; and h. Testing/PM routine.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Operate the APC system, to include: a. Describe the procedure for manual operation; and b. Describe the method of automatic operation.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Verify basic operating parameter of the Drencher fire fighting system, to include: a. Direction of flow; b. Operating pressures; and c. Activation permission parameters.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the Drencher fire fighting system, to include: a. Location of all local call points; b. Location of all remote control panels; c. Location of section valves; d. Location of drain valves; e. Location of all heat/smoke activated devices (where fitted); f. Arrangement of piping; g. Location of all spray heads; and h. Direction of flow.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Operate the Drencher figher fighting system, to include: a. Describe the procedure for manual / automatic activation b. Describe the different methods of activation; and c. Describe the procedure for Testing/PM routine.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate watertight Sliding door system by; a. Locally in the Machinery room; b. Remotely via manual pumping station; c. Remotely via HMI; and d. Locate all HMI.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

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EO 007.02 INSPECT THE SEAWATER COOLING SYSTEM IN EMERGENCY CONFIGURATION

1. Performance: Inspect the Seawater Cooling System in Emergency Configuration.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect the Seawater cooling system in emergency configuration IAW refs, to include:

    a. Verifying basic operating parameters of the bildge eductor system; and

    b. Operating the bilge eductor system.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Verify basic operating parameters of the bilge eductors to include: a. Direction of flow; b. Eductor flow rates; and c. Valve activation sequence.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the bilge eductors, to include: a. Location of mode of water; b. Location of Suction; c. Location Overboard discharge; d. Location of lazy rods/deck fitting; e. Location of Emergency overboard discharge connections; and f. Location of submersible pumps.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Operate the bilge eductor, to include: a. Normal Operation; b. Emergency Operation; c. Valve Sequence; and d. System isolation/Pm routines.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

    a. HDW vessel.

7. Remarks: Nil.

EO 007.03 INSPECT PORTABLE FIREFIGHTING EQUIPMENT

1. Performance: Inspect Portable Firefighting Equipment.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect portable firefighting equipment IAW refs, to include:

    a. Conducting preventative maintenance on the Portable Firefighting Equipment.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Identify the location of, basic functionality and relevant safety considerations for the following key components of the portable firefighting equipment, to include: a. Foam Extinguishers; b. CO2 Extinguishers; and c. Powder Extinguishers.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Conduct preventative maintenance on the portable firefighting equipment as per the manufacutre's guidelines, to include: a. Foam Extinguishers; b. CO2 Extinguishers; and c. Powder Extinguishers.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 007.04 INSPECT SEAWATER SERVICES SYSTEM

1. Performance: Inspect Seawater Services System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect seawater services system, IAW refs, to include:

    a. Verifying the basic operating of parameters of the seawater services system;

    b. Operating the Seawater Services system; and

    c. Operating the Seawater Services system in an emergency.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Verify basic operating parameters to include: a. Seawater Services working pressure (min/max); and b. Pump Management.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the seawater services system, to include: a. Location of all fire pumps; b. Location of main isolations; c. Arrangement of piping; d. Location of all hydrants; e. Direction of flow; and f. Cold weather precautions.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Operating fire pumps to include: a. Local/remote activation; b. Flash-up / shut down procedures; and c. Running checks.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operating seawater servies in an emergency, to include: a. Manual pump skid activation; b. Emergency bilge suction; c. Cross-connect with SW Cooling system; and d. Manual Drencher system activation.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 007.05 INSPECT SMOKE CLEARING HVAC SYSTEM

1. Performance: Inspect Smoke Clearing HVAC System.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall inspect smoke clearing HVAC system IAW refs, to include:

    a. Verfiying basic operating parameters; and

    b. Operating the smoke clearing HVAC system.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Verify basic operating parameters to include: a. Fan flow rate (CFM) capacities; b. Proper system configuration and operation; and c. Smoke control conditions.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.
Identify the location of, basic functionality and relevant safety considerations for the following key components of the smoke clearing HVAC system, to include: a. Machinery space configuration; b. Zone configuration; c. Location of all pneumatic valves; d. Location of all fans; e. Flying station configuration; f. Layout of trunking; g. Location of all portable fans; h. Location of all temporary trunking; i. Smoke detectors; and j. Heat detectors.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

Teaching Points	Instructional Strategies	Time	References/Remarks
Operate the smoke clearing HVAC system, to include: a. Activation methods; b. Location of local activation switches; c. Remote activation (IPMS); d. Locate operate / test CCTV; and e. System isolation/Pm routines.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A74, A75, A76, A77, A78, A79, A80, A81, A82, A83, and A84. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

PO 008 – RESPOND TO EMERGENCY CONDITIONS

EO 008.01 PERFORM AT-SEA EMERGENCY PROCEDURES

1. Performance: Perform At-Sea Emergency Procedures.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall perform at-sea emergency procedures, IAW refs, to include:

    a. Operating F/O system locally;

    b. Operating steering system locally to include:

        (1) Operating steering systems manually.

    c. Operating OG&D systems.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Conduct Engineering Drills: a. Operate FOS quick closing valves; b. Steering gear break down; c. Causualty clearing; d. Fuel spill; e. Hazmat spill; f. Loss engine room air supply; and g. Machinery space fire.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A15, A18, A30, A29, A41, A46, A47, A72, A57, A12, A13, A37, A61, A82, and D1. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

EO 008.02 RECTIFY ALARMS

1. Performance: Rectify Alarms.

2. Conditions:

    a. Given:

        (1) System schematics;

        (2) Access to systems;

        (3) Required tools;

        (4) PPE;

        (5) References; and

        (6) Supervision.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) HDW Class vessel alongside or at sea.

3. Standard: The trainee shall rectify alarms, IAW refs, to include:

    a. Configuring power supplies.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
Respond, acknowledge and reset local alarm conditions for the following: a. Steering gear alarms; b. Switchboard alarms; c. UPS alarms; d. SFC alarms; e. MDG SaCosOne LOP; f. Active shaft grounding; g. Service air h. Start air i. Chillers; j. Sewage treatment and abatement system;	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A15, A18, A30, A29, A41, A46, A47, A72, A57, A12, A13, A37, A61, A82, and D1. Remarks: Nil.
Configure power supplies, to include: a. SFC; and b. Medical UPS.	Method: OJT Media: Real Equipment Environment: HDW vessel	Conducted during OJT	Refs: A1, A2, A15, A18, A30, A29, A41, A46, A47, A72, A57, A12, A13, A37, A61, A82, and D1. Remarks: Nil.

5. Total EO Time as follows:

   a. Conducted during 3 month OJT.

6. Training and Learning Aids:

   a. HDW vessel.

7. Remarks: Nil.

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QUALIFICATION STANDARD ANNEXES

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A1	RDIMS #629406	HDW CLASS SEMS
A2		SSO (AL11)
A3		HDW CLASS FLASH UP ORDERS
A4		HDW CLASS PRE-SAIL CHECKLIST
A5		HDW HV SOP-S13
A6	RDIMS: 461906	NAVORD 11100-1 Qualifications and Access Requirements for High Voltage Ships
A7	C-24-715-000-MS-001	MAIN PROPULSION SYSTEM
A8	C-24-718-000-MS-000	PROPULSION TRANSFORMER
A9	C-24-719-000-MS	PROPULSION CONVERTER MV7315 VOLUME 1 – 9
A10	C-25-583-000-MS-001	MAIN LUBE OIL SYSTEM
A11	C-25-585-000-MS-001	LUBE OIL PURIFIER EQUIPMENT SYSTEM
A12	C-25-586-000-MS-001	FUEL OIL SERVICE SYSTEM
A13	C-25-587-000/MS-001	FUEL OIL PURIFIER SYSTEM

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A14	C-25-597-000-MS-001	LUBE OIL TRANSFER PUMP GP41EM-1U2B2
A15	C-26-541-000-MS-001	PRIMARY ELECTRICAL POWER GENERATION AND DISTRIBUTION
A16	C-26-543-000-MS	6.6KV DISTRIBUTION SYSTEM (VOLUME 1- 4)
A17	C-38-463-000-MS-001	440V DISTRIBUTION EQUIPMENT
A18	C-38-465-000-MS-001	440V SWITCHBOARDS
A19	C-38-457-000-MS-001	220V DISTRIBUTION EQUIPMENT
A20	C-26-544-000-MS	6.6kV DIESEL ENGINE ASSEMBLY
A21	C-26-545-000-MS	6.6kV ALTERNATOR (VOLUME 1-2)
A22	C-38-475-000-MS-001	EMERGENCY HARBOUR DIESEL GENERATOR
A23	C-38-486-000-MS-001	EMERGENCY HARBOUR DIESEL CAT ENGINE
A24	C-38-464-000-MS-001	440V MOTOR CONTROL CENTRES
A25	C-57-876-000-MS-001	NAVIGATION LIGHTING GROUP
A26	C-26-546-000-MS-000	NEUTRAL GROUND RESISTOR
A27	C-26-547-000-MS-001	AUTOMATIC VOLTAGE REUGLATOR
A28	C-26-551-000-MS-000	BATTERY CHARGER 110 VDC UPS
A29	C-48-355-000-MS-001	INTEGRATED PLATFORM MANAGEMENT SYSTEM (IPMS)
A30	C-38-496-000-MS-001	SHIP SERVICE UPS
A31	C-48-406-000-MS-001	NOXIOUS GAS SYSTEM

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A32	C-48-407-000-MS-001	DEADMAN SYSTEM
A33	C-26-556-000-MS	UPTAKES EXHAUST SYSTEM (VOLUME 002-005)
A34	C-26-557-000/ME-001	INSTALLATIONS/ASSEMBLY INSTRUCTIONS – EXHAUST SILENCER FOR MAN L32/44CR)
A35	C-26-567-000/ME-001	INSTALLATIONS/ASSEMBLY INSTRUCTIONS – MDT MEDIUM SCR SYSTEM
A36	C-26-583-000-MS-000	SHIP SERVICE TRANSFORMER (VOLUME 1-3)
A37	C-27-G15-000-MS	FUEL OIL SYSTEMS (VOLUME 1-3)
A38	C-27-G17-000-MS	QUICK CLOSE VALVE ACTUATION SYSTEM (VOLUME 1-2)
A39	C-27-G19-000-MS-001	EQUIPMENT COOLING SYSTEMS
A40	C-77-411-000-MS-001	SEAWATER SERVICE SYSTEM (VOLUME 1-2)
A41	C-27-G20-000-MS	ANTI-FOULING SYSTEM (VOLUME 1-2)
A42	C-38-442-000-MS-001	CATHODIC PROTECTION SYSTEM
A43	C-27-G24-000-MS	THERMAL OIL SYSTEM (VOLUME 1-3)
A44	C-27-G73-000-MS-001	THERMAL OIL CONTROL PANEL
A45	C-27-G25-000	TANK HEATING SYSTEM
A46	C-27-G28-000-MS-001	COMPRESSED SERVICE AIR SYSTEM
A47	C-27-H41-000-MS-001	STARTING AIR COMPRESSOR
A48	C-27-G29-000-MS	BALLAST SYSTEM (VOLUME 1-2)
A49	C-27-G78-000-MS-001	BALLAST/BILGE PUMP S100-80-275N D02

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A50	C-27-G30-000-MS-001	HEELING SYSTEM
A51	C-27-G31-000-MS-001	ACTIVE ROLL FIN STABILIZATION SYSTEM
A52	C-27-G32-000-MS-001	DESALINATION SYSTEM
A53	C-27-G33-000-MS	COLD FRESH WATER SYSTEM (VOLUME 1-5)
A54	C-27-G34-000-MS-001	HOT FRESH WATER SYSTEM
A55	C-27-H10-000-MS-001	ULTRAVIOLET STERILIZER
A56	C-27-G35-000-MS-001	SOLID WASTE SYSTEM
A57	C-27-G36-000-MS	SANITARY SYSTEM (VOLUME 1-3)
A58	C-27-G37-000-MS	OIL POLLUTION ABATEMENT SYSTEM (VOLUME 1-2)
A59	C-27-G38-000-MS	BALLAST WATER TREATMENT SYSTEM (VOLUME1-3)
A60	C-27-G39-000-MS-002	INCINERATOR EQUIPMENT
A61	C-27-G40-000-MS-001	STEERING GEAR AND CONTROL SYSTEM (VOLUME 1-2)
A62	C-27-G46-000-MS	BOW THRUSTER AND CONTROL SYSTEM (VOLUME 1-2)
A63	C-27-G79-000-MS	BOW THRUSTER CONVERTER (VOLUME 1-2)
A64	C-27-G57-000-MS	BLACK & GREY WATER COLLECTION SYSTEM (VOLUME 1-2)
A65	C-27-H27-000-MS-001	GREASE SEPARATOR UNIT DVZ-FT-100 AGD
A66	C-27-G68-000-MS-001	ECONOMIZER HEAT EXCHANGER
A67	C-27-H06-000-MS-001	FRESH WATER CALORIFIER

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A68	C-28-858-000-MS-001	REPLENISHMENT AT SEA SYSTEMS
A69	C-28-913-000-MS-001	WATERTIGHT SLIDING DOORS
A70	C-29-614-000-MS	REFRIGERATION SYSTEMS (VOLUME 1-2)
A71	C-29-615-000-MS-001	HEATING VENTILATION AND AIR CONDITIONING SYSTEM
A72	C-29-631-000-MS-001	AC CHILLERS
A73	C-29-637-000-MS	HVAC FAN EQUIPMENT GROUP ENGINE ROOMS/ACCOMODATION (VOLUME 1-2)
A74	C-77-412-000-MS	FIXED FIRE EXTINGUISHING SYSTEMS (VOLUME 1-2)
A75	C-77-419-000-MS-001	AQUEOUS FILM FORMING FOAM (AFFF)
A76	C-77-420-000-MS-001	WATER MIST SYSTEM
A77	C-77-421-000-MS-001	FIXED GAS SYSTEM (NOVEC)
A78	C-77-429-000-MS-001	BILGE SYSTEM
A79	C-77-432-000-MS-001	FIXED DETECTION SUPPRESSION AND NOTIFICATION SYSTEM
A80	C-77-438-000-MS-001	FIRE PUMP
A81	C-77-439-000-MS-001	FIRE CHARGE PUMP
A82	CFCD132	RCN DAMAGE CONTROL MANUAL
A83	C-23-HDW-000/MS-003	MANUAL OF TRIM AND STABILITY HDW CLASS
A84		SHOPS (FLYING STATIONS)
A85	C-77-417-000	DRENCHER
A86	27-G82-000	TUNNEL THRUSTER

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A87	24-717-000_	SHAFTING AND PROPELLER
A88	C-28-851-000-MS-002	GASOLINE (F67) FUEL SYSTEM, FREE STANDING GAS TANK AND FOUNDATION
A89	C-03-050-HDW-MB-001	DRAFT-Manual of Emergency Procedures MARINE SYSTEMS ENGINEERING (AT SEA)
A90	C-24-731-000-MB-001	STERN TUBE SEAL
A91	C-24-741-000-MB-001	WATER QUALITY PACKAGE
B		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
B01		
B02		
B03		
C		**Civilian Publications and Manuals that relate to this Qualification**
C01		
C02		
D		**Other Publications and Manuals that relate to this Qualification**
D1	AOPS PSMR* FMEA Operation Principles	Propulsion and Steering Machinery Redundancy (PSMR*)
D2		OFFICER IN CHARGE OF ENGINEERING WATCH MODEL - 7.04
D3		STCW CONVENTION AND STCW CODE 2017 EDITION

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
D4	C-014006 / P21023856_0011_MCM_AOPS SMPT IEPP	CANADA AOPS SYNTHETIC MAINTENANCE PROCEDURAL TRAINERS (SMPT) MAINTAINER COURSE MATERIAL (MCM)

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Note for all Tasks as this QSP without an approved Work Requirement Table the task listed in the performance statements will not directly align with the task listed below. But both the task from the WRT and the task listed in the performance statement should be fairly similar and describe the same onboard activities.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
AT2620	Verify the Fluid Level In a Tank	T	005	
AT2610	Verify the Correct Operation of Crash on Deck Alarm	T	004	
AT2600	Verify the Correct Operation of a Man Overboard Alarm System	T	004	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
AT2590	Verify the Correct Operation of a Man Locked In Fridge Alarm System	T	004	
AT1970	Start / Stop an Oil Pollution Abatement System	T	003	
AT1960	Start / Stop an Line Handling Winch	T	004	
AT1900	Start / Stop an Auxiliary Seawater Circulation Pump	T	007	
AT1880	Start / Stop a Ventilation System	T	006	
AT1890	Start / Stop a Ventilation / Exhaust System	T	006	
AT1870	Start / Stop a Urea System	T	002	
DT3630	Start / Stop a Turning Gear Motor	T	001	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
AT1860	Start / Stop a Trim, Bilge and Ballast System	T	003	
AT1840	Start / Stop a Steering System	T	003	
AT1820	Start / Stop a Static Frequency Converter (SFC)	T	008	
AT1800	Start / Stop a Seawater Service System Pump (Firemain)	T	007	
AT1790	Start / Stop a Reverse Osmosis Desalination Unit (ROD)	T	007	
AT1780	Start / Stop a Refrigeration System	T	006	
AT1750	Start / Stop a Motor Generator	T	004	
AT1730	Start / Stop a Main Seawater Circulation System	T	007	
AT1700	Start / Stop a Main Lube Oil Filling and Transfer System	T	003	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
AT1660	Start / Stop a LP Air Compressor	T	003	
3	Start / Stop a Hydraulic Plant	T	003	
AT1580	Start / Stop a HP Air Compressor	T	003	
AT1560	Start / Stop a Heeling System	T	003	
AT1500	Start / Stop a Fuel Oil Service System	T	003	
AT1480	Start / Stop a Fuel Filling and Transfer System	T	003	
AT1460	Start / Stop a Fresh Water Cooling System	T	003	
AT1440	Start / Stop a Fin Stabilizer System	T	003	
AT1420	Start / Stop a Diesel Engine	T	002	
AT1350	Start / Stop a Calorifier	T	003	
AT1340	Start / Stop a Black & Grey Water	T	003	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
	Treatment System			
AT1330	Start / Stop a Black & Grey Water Collection System	T	003	
AT1320	Start / Stop a Bilge Pumping System	T	007	
AT1310	Start / Stop a Battery Charging System	T	005	
2	Monitor Mechanical Equipment	T	001-003	
AT1170	Monitor Electrical Equipment	T	004	
AT1150	Monitor an Electrical System	T	004	
AT1140	Monitor a Pneumatic System	T	007	
AT1120	Monitor a Hydraulic System	T	003	
AT1110	Monitor a Fuel Vent - Minimize	T	008	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
	the Impact of a Fuel Spill			
AT1090	Monitor a CCTV System	T	007	
AT0320	Isolate / Restore Power To Equipment or a System During Emergencies	T	008	
AT0280	Engage / Disengage a Turning Gear	T	001	
AT0120	Clean Up Fuel Spills and Leaks	T	008	

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ANNEX C – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title	USQ – Roundsperons HARRY DEWOLF Class
Location of Board	NFS(P) WP1020 Haida Bldg
Dates	1-5 November 2022

Board Composition

	Surname	Initials	Rank	Unit
Board Chair	MacMillan	C	CPO2	NFS(P)
TDO (Secretary)	Timbers	A	Lt	NFS(P)
TM	Sutton	J	CPO2	NTG HQ
SME	Gagnon	P	CPO2	PCC(A)
SME	Reid	A	CPO2	NFS(P)
SME	Ablett	R	MS	NFS(A)
SME	Brown	M	MS	NFS(A)

Day #	Details	Action
Pre-board	<u>Board Objective</u>: As this is the initial board conducted for the AOPV Roundsperson, the objective will be to take the existing draft If one NFS has training aids the other does not, we will make changes based on the school with less effective training aids.	

Day #	Details	Action
1	Initial introductions and started work on POs - Confirmed purpose of board is to review document created during ICT and ensure QSP meets needs of the Fleet and is within fleet school's capabilities to deliver. - Working off provisional QSP created during ICT. PO 001 - Confirmed PO content and then moved into individual EOs, minor secretarial changes made to PO and content added to teaching point table in EO. - Initial coaching/mentoring represents EC assessment as this person identifies when the trainee has demonstrated that standard has been met. Trainee is then assessed by more senior tech as the PC to ensure the standard has been met. PO 002 - Content added and removed teaching point table to accurately reflect the job. - Ongoing discussion with regards to assessment and how to accurately describe standard needed and how ensure objectivity. PO 003 Discussion on use of describe, it is the responsibility of the roundsperson to describe the operation/status of an engine, etc. to the watch-keeper. Task is to provide watch-keeper a description so they can make executive decision on what COA to pursue.	
2	Revisited PO 001 added content "inspect main propulsion converter"	

Day #	Details	Action
	PO 003 - Restarted work on PO/EO and teaching points - EO 003.07 Describe the operation of the ship's service. S3 is expected to describe current operating status of component X as part of turnover to next watch. Navigate: to get to the correct screen/view on instrument terminal to monitor applicable component/system.	
3	Continued working on PO 003 - Clear with SMEs to replace conduct shipboard tour with conduct rounds on. Trainee will still be assessed by a shipboard tour with a more snr. Tech. - Refined teaching points to more accurately reflect rounds person responsibility Started work on PO 004 - Remove EO 004.04 Inspect Batteries as task is less involved as it was in the past (all batteries are closed system now). Remaining teaching points are move to EO 003.04, EO 003.05, 004.05 (now EO 004.04) and EO 004.08 (now EO 004.08) - Teaching points removed from pre sail check as these knowledges and skills are have already been covered in previous PO/EOs what remained was to instruct the procedure itself and associated documentation. Completed EO 005 and EO 006.	
4	Continued refining teaching points	

Day #	Details	Action
	- EO 007.01 Firefighting was moved to Ships tour as it already covered by the duty watch training and combat readiness training - EO 007.03 Inspect portable firefighting equipment is the responsibility of the maintenance watch not roundsperson PO 008 At this point the point the only response for emergencies is to report to the MCR and await further direction the skills and knowledges required to complete task they could be directed to complete have already been covered.	
Post Board	Significant work was done after the board sat to further refine the QSP to align train	

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted QS Change Request (QSCR), thus ensuring the QSP has the most relevant information.

Version:	**X.X**
Date:	**Who Initiated Change:**
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	

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Trg Mgr:

TRAINING PLAN ANNEXES

ANNEX D – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.				EO001.02 and 009.11 Security Classification required
b.				
c.				
NOTES:				

3. Ammunition

SER	LOG GUIDE	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

4. Weapons

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				

NOTES:		

5. Specialized/Technical Equipment

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				

NOTES:		

6. Transport/Vehicles

SER		NSN	DESCRIPTION		QTY	REMARKS
a.		XXX-XX-XXX-XXXX				
b.		XXX-XX-XXX-XXXX				
c.		XXX-XX-XXX-XXXX				

	NOTES:		

7. Communication Equipment/Frequencies

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				

NOTES:		

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8. Training Aids

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				

NOTES:		

9. POL Products

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				

NOTES:		

10. General Purpose Stores

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				

NOTES:		

11. Stationery

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				

NOTES:		

12. Personal Equipment

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

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ANNEX E – COSTING INFORMATION

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: EXAMPLE		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	14	$1,000.0		$14,000	$14,000	$70,000.00
R&Q	14	$36.0	20	$10,080	$10,080	$50,400.00
Incidentals	14	$17.5	20	$4,900	$4,900	$24,500.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	16	$1,000.0	1	$16,000	$0	$16,000.00
IT Equip (Projector, Smart board etc.)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc.)	15	$30.0	1	$450	$450	$2,250.00

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Subtotal				**$28,265**	**$1,265**	**$33,325.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	2	$223.0	20	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				
		subtotal		**$8,920**	**$8,920**	**$44,600.00**
	Total Estimated Costs			**$66,165**	**$39,165**	**$222,825**

ANNEX F – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NTORD 4500-22 Annex B (QSP Instructions).

2. The entiretly of this course is a three month OJT. All training and assessment activities are conducted during the three months. Extension can be granted IAW NTORD 4500-55 On-the-Job Training Management.

3. There is no prescribed training sequence for these QSP as the training sequence and timings are dependent on the operational tempo of the vessel conducting the training.

ANNEX G – ASSESSMENT PLAN

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
PC 001	PO 001	Perform Main Propulsion System Rounds	Signature Block in Appendix 1	During OJT	During OJT	Functional HDW Class Vessel	Nil	IAW Annex I	TRB IAW NTORD 4500-54
PC 002	PO 002	Perform Primary PG&D System Rounds	Signature Block in Appendix 1	During OJT	During OJT	Functional HDW Class Vessel	Nil	IAW Annex I	TRB IAW NTORD 4500-54
PC 003	PO 003	Perform Marine Engineering Auxiliry Rounds	Signature Block in Appendix 1	During OJT	During OJT	Functional HDW Class Vessel	Nil	IAW Annex I	TRB IAW NTORD 4500-54
PC 004	PO 004	Perform Secondary PG&D	Signature Block in Appendix 1	During OJT	During OJT	Functional HDW	Nil	IAW Annex I	TRB IAW NTORD 4500-54

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
		System Rounds				Class Vessel			
PC 005	PO 005	Perform Deck and Hull Equipment Rounds	Signature Block in Appendix 1	During OJT	During OJT	Functional HDW Class Vessel	Nil	IAW Annex I	TRB IAW NTORD 4500-54
PC 006	PO 006	Perform Main Refrideration and HVAC System Rounds	Signature Block in Appendix 1	During OJT	During OJT	Functional HDW Class Vessel	Nil	IAW Annex J	TRB IAW NTORD 4500-54
PC 007	PO 007	Perform Damage Control System Rounds	Signature Block in Appendix 1	During OJT	During OJT	Functional HDW Class Vessel	Nil	IAW Annex J	TRB IAW NTORD 4500-54

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
PC 008	PO 008	Respond to Emergency Condition	Signature Block in Appendix 1	During OJT	During OJT	Functional HDW Class Vessel	Nil	IAW Annex J	TRB IAW NTORD 4500-54

ANNEX H – ASSESSMENT CHECKLISTS

Not currently used – Do not delete

PERFORMANCE/ENABLING CHECK

PC/EC Time: ### mins (concurrent/per trainee) + ## min debrief

ASSESSMENT ACTIVITY: (Brief description of the assessment activity that students will do IOT prove they have met the Standard of the PO/EO).

Pass/fail criteria: (identify success criteria)

ASSESSMENT RESOURCES:

- (Bulleted list of everything given to trainees for PC/EC)

IMPACT OF FAILURE: Remedial Training / Re-test

(Course Name) – PC/EC XXX.XX Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

INVIGILATOR NOTES:

1. PASS / FAIL CRITERIA: Identify success criteria.

2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	**PASS**	**FAIL**	**COMMENTS**
1. Criteria			
(Success criteria taken from PO/EO standard)			

ITEM	PASS	FAIL	COMMENTS
2.			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ANNEX I – ON JOB PERFORMANCE RECORD (OJPR)
(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 1.0	DD / MM/ YYYY	New OJPR

Name and Initials	Rank	Service Number

Date Training Started	DD MMM YYYY
Date of Completion	DD MMM YYYY

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

5. See Appendix 1 for OJPR sign off list. The remarks column in Appendix 1 shall be used for the following information only:

a. The unit in which the training and assessment were carried out; and

b. Any unusual circumstances which preclude compliance with an objective.

6. Appendix 1 Performance Checks require the trainee to demonstrate the listed competencies as described in chapter 4. A series of questions will be asked from each EO to satisfy the knowledge of the roundsperson before granting qualification. The trainee will have three (3) months from being posted to an operational ship to complete Appendix 1.

7. On completion of this OJPR, the NTD Training Coordinator shall email (scanned) the Certificate of Qualification for MITE update to:

a. +NFS(A) MITE@NFS(A)@Halifax; or

b. NFS(P) MITE@NFS(P)@Esquimalt.

8. If you wish to make a change, send your request to:

a. +NTG MITE@NTG HQ@Esquimalt

PROGRESS REVIEW

9. Progress review shall be carried out monthly by the Snr Mechanical Propulsion Manager and Snr Electrical Propulsion Manager.

OJPR Reviewed By (Rank, Position, Name)	Signature	Date

TRAINING COMPLETION CERTIFICATE

Date training completed:	
Snr Mechanical Propulsion Manager (Print and Sign)	Date:

Snr Electrical Propulsion Manager (Print and Sign)	Date:

CERTIFICATION OF QUALIFICATION

I certify that _____ completed training to the standards stated in the Marine Technician Basic Maintainer Harry DeWolf Class Package, except as noted. The member is qualified to carry out the duties of a Marine Technician Engineering Roundsperson and is ALM* qualified.

Naval Technical Officer (Print and Sign)	Date

APPENDIX 1

PO/EO	Description	Initial Assessment	Date	Signature	Remarks
PO 001 Perform Main Propulsion System Rounds (IAW TP's in Chapter 4)					
EC 001.01	**Inspect Shafting and Propeller Systems**				
	a. Inspect the Propulsion Motors.		Initial: Final:		
	b. Inspect the Sterntube Quality Water Package.		Initial: Final:		
	c. Inspect Converter Cooling Cabinet		Initial: Final:		
	d. Inspect Converter Control Cabinet		Initial: Final:		
	e. Inspect Propulsion Transformer Cooling Fans		Initial: Final:		
PO 002 Perform Primary PG&D Rounds (IAW TP's in Chapter 4)					
EC 002.01	**Inspect MAN Diesel Engines**				
	a. Deliver a shipboard tour of MAN Diesel.		Initial: Final:		
	b. Explain operation of MAN Diesel.		Initial: Final:		
EC 002.02	**Inspect Exhaust Systems**				
	a. Deliver a shipboard tour of the Exhaust Systems.		Initial: Final:		

	b. Deliver a shipboard tour of the Selective Catalytic Reduction system.		Initial: Final:		
	c. Operate the Selective Catalytic Reduction (SCR) system.		Initial: Final:		
PO 003 Perform Marine Engineering Auxiliary Rounds (IAW TP's in Chapter 4)					
EC 003.01	**Inspect/Operate Steering System.**				
	a. Describe the operation of the steering system		Initial: Final:		
	b. Deliver a tour of the steering system		Initial: Final:		
	c. Operate the steering system locally		Initial: Final:		
EC 003.02	**Inspect the Bow Thruster and Systems.**				
	a. Delivering a shipboard tour of the Bow Thruster.		Initial: Final:		
	b. Conducting pre-start checks of the Bow Thruster.		Initial: Final:		
	c. Operating locally the Bow thruster		Initial: Final:		
EC 003.03	**Inspect the FOS System.**				
	a. Describe theory of operation of the FOS system.		Initial: Final:		
	b. Deliver shipboard tour of the FOS system.		Initial: Final:		

	c. Operate the FOS system locally.		Initial: Final:		
EC 003.04	**Inspect the FOT System.**				
	a. Describe theory of operation of the FOT system.		Initial: Final:		
	b. Deliver shipboard tour of operation of the FOT system.		Initial: Final:		
	c. Operate the FOT system locally.		Initial: Final:		
EC 003.05	**Inspect Fuel/Lube Oil Purifiers.**				
	a. Describe theory of operation of the Fuel/Lube Oil Purifiers.		Initial: Final:		
	b. Deliver shipboard tour of the Fuel Oil Purifier.		Initial: Final:		
	c. Operate the Fuel Oil Purifiers locally.		Initial: Final:		
	d. Deliver a shipboard tour of the Lube Oil Purifier.		Initial: Final:		
	e. Operate the Lube Oil Purifiers locally.		Initial: Final:		
EC 003.06	**Inspect Diesel Lube Oil Fill/Transfer System.**				
	a. Deliver a shipboard tour of the DLO system.		Initial: Final:		
	b. Operate the DLO system.		Initial: Final:		

	c. Deliver a shipboard tour of the DLO Purifier system.		Initial: Final:		
	d. Operate the DLO Purifier system.		Initial: Final:		
EC 003.06	**Inspect Compressed Air Systems.**				
	Inspect Ship's Service Air:				
	a. Describing the operation of the Ship's Service (SS), Ship's Start (SST) and Breathing Air systems		Initial: Final:		
	b. Delivering a shipboard tour of the SS Air system		Initial: Final:		
	c. Operating the SS Air system locally		Initial: Final:		
	Inspect Ship's Start Air:				
	a. Delivering a shipboard tour of the SST Air system		Initial: Final:		
	b. Operating the SST Air system locally		Initial: Final:		
	Inspect Breathing Air Compressors:				
	a. Delivering a shipboard tour of the Breathing Air Compressors		Initial: Final:		
	b. Operating the Breathing Air Compressor locally		Initial: Final:		
EC 003.08	**Inspect Fresh Water Systems.**				
	Inspect Cold Fresh Water:				

	a. Describe the operation of the Fresh Water system.		Initial: Final:		
	b. Deliver a shipboard tour of the Fresh Water system.		Initial: Final:		
	c. Operate the Fresh Water Services system locally.		Initial: Final:		
	Inspect Hot Fresh Water:				
	a. Deliver a shipboard tour of the Hot Fresh Water system.		Initial: Final:		
	b. Operate locally the Hot Fresh Water system.		Initial: Final:		
	Inspect the Fresh Water Purification System:				
	a. Deliver a shipboard tour of the FW Purification system.		Initial: Final:		
	b. Operate the FW Purification system.		Initial: Final:		
	Inspect the Reverse Osmosis Desalination (ROD) System:				
	a. Deliver a shipboard tour of the ROD.		Initial: Final:		
	b. Operate locally the ROD.		Initial: Final:		
EC 003.09	**Inspect Black and Grey Water Collection/Treatment Systems.**				
	a. Describing the operation of the Black/Grey Water system		Initial: Final:		
	Inspect the Black Water (BW) system:				

	a. Deliver a shipboard tour of the BW system.		Initial: Final:		
	b. Operate the BW system		Initial: Final:		
	Inspect the Grey Water (GW) system:				
	a. Deliver a shipboard tour of the GW system		Initial: Final:		
	b. Operate the GW system		Initial: Final:		
EC 003.10	**Inspect the Oil Pollution and Abatement System.**				
	a. Deliver a shipboard tour of the OPA system.	.	Initial: Final:		
	b. Operate the OPA system		Initial: Final:		
EC 003.11	**Inspect the Ballast Water Treatment System.**				
	a. Describe the operation of the BWTS.		Initial: Final:		
	b. Deliver a tour of the BWTS.		Initial: Final:		
	c. Operate the BWTS locally.		Initial: Final:		
EC 003.12	**Inspect Ballast Water System.**				
	a. Describe the operation of the BWS.		Initial: Final:		

	b. Deliver a tour of the BWS.		Initial: Final:		
	c. Operate the BWS.		Initial: Final:		
EC 003.13	**Inspect the Incinerator System.**				
	a. Describe the operation of the Incinerator system		Initial: Final:		
	b. Deliver a tour of the Incinerator system		Initial: Final:		
	c. Operate the Incinerator system		Initial: Final:		
EC 003.14	**Inspect Equipment Cooling Systems.**				
	Inspect the Central Fresh Water Cooling (CFWC) system:				
	a. Describe the operation of the CFWC system.		Initial: Final:		
	b. Deliver a shipboard tour of the CFWC system		Initial: Final:		
	c. Operate the CFWC system		Initial: Final:		
	Inspect the Seawater Feed/Cooling (SWC) system:				
	a. Describe the operation of the SWC system.		Initial: Final:		
	b. Deliver a shipboard tour of the SWC system.		Initial: Final:		

	c. Operate the SWC system.		Initial: Final:		
EC 003.15	**Inspect the Thermal Oil System.**				
	a. Describe the operation of the Thermal Oil system.		Initial: Final:		
	b. Deliver a tour of the Thermal Oil system.		Initial: Final:		
	c. Operate the Thermal Oil system locally.		Initial: Final:		
EC 003.16	**Inspect the Tank Heating System.**				
	a. Describe the operation of the Tank Heating system		Initial: Final:		
	b. Deliver a tour of the Tank Heating system		Initial: Final:		
	c. Operate the Tank Heating system locally		Initial: Final:		
EC 003.17	**Inspect the Stability System.**				
	Inspect Active Roll Fin Stabilization system:				
	a. Describe the operation of the Active Fin Roll Stabilization system.		Initial: Final:		
	b. Deliver a tour of the Active Fin Roll Stabilization system.		Initial: Final:		
	c. Operate the Active Fin Roll Stabilization system locally.		Initial: Final:		

	Inspecting Heeling system:				
	a. Describe the operation of the Heeling system.		Initial: Final:		
	b. Deliver a tour of the Heeling system.		Initial: Final:		
	c. Operate the Heeling system locally.		Initial: Final:		
PO 004 Perform Secondary PG&D Rounds					
EC 004.01	**Inspect Secondary PG&D.**				
	a. Observing safety precautions when working with electricity.		Initial: Final:		
	b. Describe the operation of the Secondary PG&D.		Initial: Final:		
	c. Deliver a shipboard tour of the Secondary PG&D.		Initial: Final:		
	d. Operate Secondary PG&D systems.		Initial: Final:		
	e. Operate electrical LV switchboard remotely.		Initial: Final:		
	f. Test UPS.		Initial: Final:		
	g. Test SFC.		Initial: Final:		
EC 004.02	**Inspect Electrical Systems.**				

	a. Describe the operation of inspecting Electrical systems.		Initial: Final:		
	b. Deliver a shipboard tour in the inspection of electrical systems.		Initial: Final:		
	c. Perform the inspection of Electrical systems		Initial: Final:		
	d. Operate Ship's Whistle.		Initial: Final:		
	e. Test shipboard alarms.		Initial: Final:		
	f. Inspect whistle assemblies.		Initial: Final:		
	g. Inspect whistle at will switches.		Initial: Final:		
	h. Operate Integrated Platform Management System.		Initial: Final:		
	i. Operate Navigation lights locally.		Initial: Final:		
EC 004.03	**Inspect Cathodic Protection System.**				
	a. Operate the Cathodic Rectifiers.		Initial: Final:		
	b. Check status of shaft grounding system.		Initial: Final:		
	c. Check system potential and current meters.		Initial:		

			Final:		
	d. Check condition of shaft grounding brush gear and slip rings.		Initial: Final:		
	e. Operate cathodic protection system.		Initial: Final:		
	f. Operating portable reference electrode.		Initial: Final:		
EC 004.05	**Inspect Small Boat Electrics.**				
	a. Apply PPE protocols.		Initial: Final:		
	b. Inspect small boat electrics.		Initial: Final:		
EC 004.06	**Operate Deck Machinery.**				
	a. Operate the winch power units in all modes.		Initial: Final:		
	b. Operate the capstans and anchor windlass.		Initial: Final:		
EC 004.07	**Conduct Electrical Pre-Sail Checks.**				
	a. Conduct pre-sail checklist.		Initial: Final:		
EC 004.08	**Inspect the Emergency Harbour Diesel Generator (EHDG).**				
	a. Deliver a shipboard tour of the EHDG		Initial: Final:		

	b. Operation of the EHDG		Initial: Final:		
PO 005 Perform Deck and Hull Equipment Rounds					
EC 005.01	**Inspect Auxiliary Boats.**				
	a. Conduct rounds on Lifeboats.		Initial: Final:		
	b. Conduct rounds on MRRB's.		Initial: Final:		
	c. Conduct rounds on RHIB's.		Initial: Final:		
	d. Conduct rounds on Landing Craft.		Initial: Final:		
EC 005.02	**Inspect the Gasoline (F67) Fuel System.**				
	a. Deliver a shipboard tour of the Gasoline (F67) Fuel system.		Initial: Final:		
	b. Maintain Gasoline (F67) Fuel system.		Initial: Final:		
EC 005.03	**Inspect/Maintain Lifting Appliances.**				
	a. Deliver a shipboard tour of the Davits.		Initial: Final:		
	b. Maintain the Davits.		Initial: Final:		
EC 005.04	**Inspect Liquid Handling Systems.**				

	a. Describe the operation of hull fitted tanks.		Initial: Final:		
	b. Deliver a shipboard tour of hull fitted tanks.		Initial: Final:		
	c. Operate the Liquid Level Management system.		Initial: Final:		
PO 006 Perform Main Refrigeration and HVAC Rounds					
EC 006.01	**Inspect the Main Refrigeration System.**				
	a. Describe the operation of the Main Refrigeration system.		Initial: Final:		
	b. Deliver a shipboard tour of the Main refrigeration system.		Initial: Final:		
	c. Operate the Main Refrigeration system.		Initial: Final:		
EC 006.02	**Inspect Air Conditioning and Heating Systems.**				
	Inspect a Chiller:				
	a. Deliver a shipboard tour of a Chiller.		Initial: Final:		
	b. Operate a chiller.		Initial: Final:		
	Inspect the Chilled Water system:				
	a. Deliver a shipboard tour of the Chilled Water system.		Initial: Final:		
	b. Operate the Chilled Water systems.		Initial:		

			Final:		
	Inspect the Heating Ventilation and Air Conditioning (HVAC) system:				
	a. Deliver a shipboard tour of the HVAC system.		Initial: Final:		
	b. Operate the HVAC system.		Initial: Final:		
	Inspect the Hot Water HVAC system:				
	a. Deliver a shipboard tour of the Hot Water HVAC system.		Initial: Final:		
	b. Operate the Hot Water HVAC system.		Initial: Final:		
PO 007 Perform Damage Control System Rounds.					
EC 007.01	**Inspect Fixed Fire Extinguishing systems.**				
	Inspect the Aqueous Film Forming Foam (AFFF) system:				
	a. Describe the operation of the Aqueous Film Forming Foam (AFFF) system.		Initial: Final:		
	b. Deliver a shipboard tour of the Aqueous Film Forming Foam (AFFF) system.		Initial: Final:		
	c. Operate the Aqueous Film Forming Foam (AFFF) system.		Initial: Final:		
	Inspect the Fixed Gas system:				
	a. Describe the operation of the Fixed Gas system.		Initial: Final:		

	b. Deliver a shipboard tour of the Fixed Gas system.		Initial: Final:		
	c. Operate the Fixed Gas system.		Initial: Final:		
	Inspect the Water Mist firefighting system:				
	a. Describe the operation of the Water Mist system.		Initial: Final:		
	b. Deliver a shipboard tour of the Water Mist system.		Initial: Final:		
	c. Operate the Water Mist system.		Initial: Final:		
	Inspect the Twin Agent Unit (TAU) firefighting system:				
	a. Describe the operation of the TAU system.		Initial: Final:		
	b. Deliver a shipboard tour of the TAU system.		Initial: Final:		
	c. Operate the TAU system.		Initial: Final:		
	Inspect the Galley Fitted Fire Suppression System:				
	a. Describe the operation of the Galley Fitted Fire Suppression system.		Initial: Final:		
	b. Deliver a shipboard tour of the Galley Fitted Fire Suppression system.		Initial: Final:		
	c. Operate the Galley Fitted Fire Suppression system.		Initial: Final:		

	Inspect the Drencher firefighting system:				
	a. Describe the operation of the Drencher system.		Initial: Final:		
	b. Deliver a shipboard tour of the Drencher system.		Initial: Final:		
	c. Operate the Drencher system.		Initial: Final:		
EC 007.02	**Inspect the Bilge Eductor System.**				
	a. Describe the operation of the Bilge Eductor system.		Initial: Final:		
	b. Deliver a shipboard tour of the Bilge Eductor system.		Initial: Final:		
	c. Operate the Bilge Eductor system.		Initial: Final:		
EC 007.03	**Inspect Portable Firefighting Equipment.**				
	a. Deliver a shipboard tour of the Portable Firefighting Equipment.		Initial: Final:		
	b. Conduct preventative maintenance on the Portable Firefighting Equipment.		Initial: Final:		
EC 007.04	**Inspect Seawater Service System.**				
	a. Describe the operation of the Seawater Services system.		Initial: Final:		
	b. Deliver a shipboard tour of the Seawater Services system.		Initial: Final:		

	c. Operate the Seawater Services system.		Initial: Final:		
	d. Operate the Seawater Services system in an emergency.		Initial: Final:		
EC 007.05	**Inspect Smoke Clearing HVAC System.**				
	a. Describe the operation of the Smoke Clearing HVAC system.		Initial: Final:		
	b. Deliver a shipboard tour of the Smoke Clearing HVAC system.		Initial: Final:		
	c. Operate the Smoke Clearing HVAC system.		Initial: Final:		
PO 008 Respond to Emergency Conditions					
EC 008.01	**Perform At-Sea Emergency Procedures**				
	a. Operate FOS quick closing valves;				
	b. Steering gear break down;				
	c. Causualty clearing;				
	d. Fuel or Hazmat spill;				
	e. Loss engine room air supply; and				
	f. Machinery space fire.				

EC 008.02	**Rectify Alarms**				
	a. Steering gear alarms;				
	b. Switchboard alarms;				
	c. UPS alarms;				
	d. SFC alarms;				
	e. MDG SaCosOne LOP;				
	f. Active shaft grounding;				
	g. Service air				
	h. Start air				
	i. Chillers;				
	j. Sewage treatment and abatement system				

ANNEX J – TT PLAN

EO	**COA 1** **TEL Delivery**	**COA 2** **Interim Delivery**	**Impact if TEL not implemented** - Realized benefits - E/W Coast Delta

ANNEX K – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

Acronym and Abbreviation	Term or Definition
A/C	Air Conditioner
AC	Alternate Current
AFFF	Aqueous Film Forming Foam
AOPV	Arctic Offshore Patrol Vessel
APC	Aqueous Potassium Carbonate
Bio	Biological
B/W	Black Water
BWS	Ballast Water System
BWTS	Ballast Water Treatment System
CBRN	Chemical Biological Radiological Nuclear
CBT	Computer Based Training
CCTV	Closed Circuit Television
CFM	Cubic Feet/Minute
CFWCS	Central Fresh Water Cooling System
D/C	Damage Control
DC	Direct Current
DFO	Diesel Fuel Oil
DLO	Diesel Lube Oil
EHDG	Emergency Harbour Diesel Generator
F/W	Fresh Water

FA	First Aid
FO	Fuel Oil
FOS	Fuel Oil System
FOT	Fuel Oil Transfer
G/W	Grey Water
HAZMAT	Hazardous Material
HMI	Human Machine Interface
HFW	Hot Fresh Water
H/W	Hot Water
HVAC	Heat and Ventilation System
IAW	In Accordance With
IPMS	Integrated Platform Management System
JP5	Jet Propellant 5
LLMS	Liquid Level Management System
LO	Lubrication Oil
MCOS	Manual Change Over Switch
MLO	Main Lube Oil
MRRB	Multi Role Rescue Boat
Nav	Navigation
OJPR	On Job Performance Record
OJT	On Job Training
PPE	Personal Protective Equipment
PG&D	Power Generation & Distribution

Pp	Pump
PRB	Progress Review Board
RHIB	Ridged Hull Inflatable Boat
SEMS	Shipboard Environmental Management System
SFC	Static Frequency Converter
SO	Standing Orders
S/W	Sea Water
SROD	Shipboard Reverse Osmosis Desalination
SSA	Ships Service Air
SSTA	Ships Start Air
SOP	Standard Operating Procedures
TAU	Twin Agent Unit
TEU	Twenty-Foot Equivalent Unit
TRB	Training Review Board
UPS	Uninterruptable Power Supply
UV	Ultraviolet
WHMIS	Workplace Hazardous Material Information Sheet
W/T	Water Tight Door